Antitrust, Market Power, and U.S. Macroeconomic Outcomes*

Brian Amorim Cabaço[†] and Morten O. Ravn[‡]

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July 17, 2026

Abstract

We provide a narrative record of U.S. federal antitrust cases from the mid-1950s to 2023. We document a fundamental shift in enforcement from civil non-merger and criminal cases toward civil merger cases around 1980. Focusing on publicly listed firms, we estimate significant negative abnormal returns following case filings and use them to construct a measure of antitrust activity. Estimates at the two-digit sector and aggregate levels show that higher antitrust activity is followed by stronger competition, with contrasting effects across case types. At both levels, the combined civil non-merger and criminal enforcement measure is associated with greater markup dispersion and weaker economic activity; at the aggregate level, the measure is also associated with lower R&D investment. Civil merger enforcement is associated with lower markup dispersion and lower activity in treated sectors, but with higher aggregate GDP, consumption, and R&D investment.

JEL Classification: C32, E30, L16, L40

Keywords: market power, business cycles, antitrust, dynamic responses

*We are grateful for comments from seminar and conference audiences at the Hydra 2023 Conference; the 2024 Society for Economic Dynamics Annual Conference; the 2023 Spanish Macroeconomics Network Meeting; Paris School of Economics; the University of Texas at Austin; Sciences Po; LUISS; the University of Mannheim; and IIES. We thank Alice Avenale, Marina Kotsa, and Diego Almonacid Lovera for excellent research assistance. Morten O. Ravn acknowledges financial support from ERC Project BUCCAC - DLV 8845598.

[†]Dept. of Economics, UCL. Email: brian.cabaco@eui.eu.

[‡]Dept. of Economics, UCL and the CEPR. Corresponding author. Email: m.ravn@ucl.ac.uk.

1 Introduction

Much recent empirical evidence has pointed towards increased market power of firms in the U.S. economy. There is comparatively less empirical evidence at a broader level on the impact of this trend. In this paper we aim to provide empirical evidence on this issue. We do so by exploiting the fact that the U.S. has a long history of antitrust legislation and enforcement aimed at addressing the potentially harmful effects of business concentration. Variations in antitrust activity should therefore be related to market power. We construct a dataset of the universe of federal antitrust indictments in the U.S. since the mid-1950s, and, focusing on cases involving publicly listed companies, we estimate their impact on firm valuations. From these estimates, we derive measures of antitrust activity which we utilize to estimate the relationship between antitrust, market power, and macroeconomic outcomes.

Antitrust legislation in the U.S. dates back to the Interstate Commerce Act of 1887, introduced to regulate the U.S. railway industry, which was considered to engage in unreasonable business practices. Broader legislation followed swiftly when the U.S. Congress passed the Sherman Act of 1890, which was aimed at preventing monopolization and collusion in U.S. commerce at large. In 1914, further legislation was introduced with the passing of the Clayton Act and the Federal Trade Commission Act. The Clayton Act specified a number of business practices considered anti-competitive, while the Federal Trade Commission Act established the Federal Trade Commission. These three acts, and later amendments, still provide the backbone of antitrust legislation in the U.S., while their interpretation has developed over time through the U.S. judicial system. At the federal level, the antitrust laws are enforced by the antitrust authorities, the Antitrust Division of the U.S. Department of Justice (ATR of the USDOJ) and the Federal Trade Commission (FTC). The antitrust cases pursued by these institutions are well documented and offer extensive information about the antitrust authorities' efforts to address anti-competitive behaviors in U.S. commerce.

We exploit the richness of the antitrust enforcement history to investigate the links between antitrust, competition, and economic outcomes. We collect data on the universe of antitrust cases launched by the Federal Trade Commission and by the U.S. Department of Justice for sample periods ending in 2023 and starting in 1954 and 1957, respectively. In this sample period, more than 6,300 cases were launched by the FTC and the USDOJ, involving close to 16,000 defendants, of which more than 11,500 were corporate firms. We broadly divide the antitrust cases into civil non-merger and criminal cases on the one hand, and civil merger cases on the other hand. The former include cases related to Sherman Act Section 1 violations such as price fixing, bid rigging, market and customer allocations, boycotts, tying, and restraints on supply; Sherman Act Section 2 violations related to monopolization; violations of Clayton Act Section 2 (price discrimination), Section 3 (tying and exclusivity), and Section 8 (interlocking boards); and those specified in later amendments. Civil merger and acquisition cases relate to violations of Section 7 of the Clayton Act and the Hart–Scott–Rodino Antitrust Improvements Act of 1976, but also to certain violations of Sections 1 and 2 of the Sherman Act and Section 5 of the Federal Trade Commission Act. More broadly, civil non-merger and criminal cases relate to the use of unfair business practices, while mergers and acquisitions relate to the formation of firms or asset acquisitions that "substantially lessen competition or tend to create a monopoly."

An interesting finding from our record of antitrust activity is that over the sample period of more than 60 years, there has been a fundamental change in antitrust focus, with a significant shift away from civil non-merger and criminal cases towards civil merger and acquisition cases. This change occurs around 1980 and is particularly evident for publicly listed firms. Another interesting fact is that publicly listed firms indicted for antitrust violations are large relative to other publicly listed firms and are concentrated among the top three deciles of the size distribution for such public firms, whether measured by sales, employees, or assets.

We exploit the data on antitrust violations to build an indicator of antitrust activity which we relate to market power and outcomes either at the sector level or at the level of the aggregate economy. The focus on outcomes at a broader level implies that we cannot study each of the antitrust cases with the richness of analysis applied in the empirical industrial organization literature. Instead, we focus on bringing out results at a more aggregated level by taking advantage of the variation in antitrust enforcement across sectors and time. We carry out our analysis in two steps. First, we estimate how antitrust indictments affect firm valuations and use those estimates of abnormal returns on indicted firms' equity in windows around the case opening dates to construct a measure of antitrust activity. We then relate these antitrust indicators to outcomes of the U.S. economy at the sector level and at the aggregate level.

Our estimates of the abnormal returns on firm equity due to antitrust indictments exploit a high-frequency event-study approach. For each indicted firm, we first estimate a Fama-French three-factor model of expected returns using data for sample periods prior to the indictments. Abnormal returns are then measured as realized returns less expected returns in windows around the case opening dates. Our results indicate highly significant negative excess returns on the case opening days as well as on the subsequent trading day. We also find some evidence of negative returns two days ahead of the case opening days. In terms of cumulated excess returns, we find that civil non-merger and criminal cases on average induce a negative excess return of around one percent at forecast horizons from 50 to 60 trading days, while at these horizons, the average cumulated excess returns for civil merger cases are around -2 percent. Given the size of our sample—more than 1,200 firms for each type of offense—the estimates of the cumulated excess returns are highly statistically significant despite variation across cases. From these estimates, we derive measures of antitrust activity by summing the implied impact on the firms' market valuations across firms, weighted by sampling uncertainty, and normalizing these measures by the total market capitalization of U.S. public firms.

We then relate the antitrust activity measures to indicators of competition, activity, investment, productivity, and markup dispersion at the two-digit sector level. For this purpose, we estimate dynamic responses to antitrust activity on the basis of panel local projections. Focusing on the two-digit level, we find that higher antitrust activity induces a decline in indicators of market power such as markups and market concentration (Herfindahl indices), regardless of the type of antitrust violation on which we focus. Thus, when a sector is more heavily scrutinized for anti-competitive behavior, market power declines as intended. In the short run, we also find that both types of antitrust indicators induce a decline in the treated sectors' real output and productivity (as measured by estimates of TFPR). However, we find contrasting effects of the two types of antitrust violations on other outcomes. In particular, we find that in response to more intense antitrust activity related to civil non-merger and criminal cases, capital expenditures of publicly listed firms in the relevant sector decline relative to other sectors, while markup dispersion rises. In contrast, higher antitrust activity related to civil merger cases spurs capital expenditures and leads to a decline in markup dispersion. We show that the results are robust to alternative measures of markups, to excluding the early part of the sample, which displayed unusually high antitrust activity, and to excluding sectors such as Finance and Real Estate or Construction, for which competition measures may be harder to estimate. These results are therefore indicative of the pro-competitive effects of civil-merger-related antitrust indictments, which also spur investment and generate a decline in misallocation, while civil non-merger and criminal cases appear to deter investment and worsen misallocation while stimulating competitive pressures.

An important concern is that the sector-level results hide spillovers and general equilibrium effects due to the "missing intercept." For that reason, we then study aggregate outcomes on the basis of local projections by aggregating the antitrust measures across sectors. In this case, we study the impact on a rich set of outcomes such as real GDP, consumption and investment, R&D spending, the real wage, unemployment, and average labor productivity. Consistent with the sector-level outcomes, we find that increased antitrust activity gives rise to a decline in markups and real corporate profits, thus indicating that antitrust promotes competition. However, the contrast between the impact of antitrust activity in terms of civil non-merger and criminal cases and the impact of civil mergers is even more evident. We find that the latter are associated with a persistent rise in aggregate real GDP, consumption, and investment, and with a stimulus to R&D spending and productivity, whether measured by TFPR or average labor productivity. Accompanying these effects, we also find a rise in real wages and a temporary decline in unemployment, alongside a short-lived decline in overall markup dispersion. For civil non-merger and criminal cases, we instead find a decline in real activity, consumption, investment, and R&D spending, while unemployment rises and real wages decline. Moreover, for this type of antitrust violation, we estimate a persistent rise in markup dispersion and a significant decline in productivity. We show that these results are robust to measurement issues related to estimates of markups and productivity, and we also investigate robustness with respect to the frequency of observation. Regarding the latter, all our baseline estimates are based on annual data, since we estimate markups and TFPR from Compustat data. An alternative is to proxy markups by the inverse of the labor share of income, in which case we estimate aggregate outcomes from quarterly data. One issue that arises when examining quarterly data is that we cannot control for misallocation through markup dispersion or for market concentration. Nonetheless, we show that many of our results are robust when studying the impact of antitrust using quarterly data.

1.1 Related Literature

A number of papers have studied trends in the number of antitrust cases pursued by the FTC and/or the USDOJ over time; see, e.g., Posner (1970), Gallo et al. (2000), or Ghosal (2011). The classic study of Posner (1970) is perhaps the most complete of these studies, covering the USDOJ (for the sample period 1890–1969), the FTC (1915–1954), and private antitrust cases (1890–1969). Gallo et al. (2000) instead focus on the USDOJ and record antitrust cases for the 1955–1997 sample. Our study thus adds 26 years of data on antitrust cases raised by the USDOJ and 69 years to the data on the FTC.

Firm-by-firm event-study estimators have been applied previously in the antitrust literature. Bosch and Eckard (1991) use a similar approach to estimate stock market responses to 127 USDOJ indictments for price fixing under Section 1 of the Sherman Act during the 1962–1980 sample period.[1] Aguzzoni, Langus and Motta (2013) and Günster and van Dijk (2016) study the impact of "dawn raids" (surprise inspections) and infringement decisions of the European Competition Commission and also adopt an event-study approach to estimate their impact on firm valuations. Our analysis studies a *much* larger set of antitrust cases over a longer sample period. Moreover, an important difference from the latter two studies is that the U.S. offers a longer history of antitrust regulation and that the success rates of the U.S. antitrust authorities are higher than those of the European Commission in its early years.[2]

A number of studies have found that various indicators of market power have risen over recent decades. This includes studies of markups estimated using the "production function approach," such as Hall (2018), Traina (2018), Hasenzagl and Pérez (2026), and perhaps most prominently De Loecker et al. (2020). Other studies have instead used a "demand-system" approach to generate estimates of markups at the broader economic level, e.g., Döpper et

[1] Bittlingmayer (1992) instead relates movements in the Dow stock price index to the number of antitrust cases raised by the USDOJ during 1904–1945. He finds that each case filed has a large impact on the Dow.

[2] Event-study approaches are, of course, used in many other applications; see, e.g., Acemoglu et al. (2016), who estimate the impact of political connections on firm performance.

al. (2025). Each of these studies documents rises in market power over the last few decades, although estimates of the size of this change, and of markups themselves, differ across studies. One shared finding of these papers is that the rise in market power as indicated by markups has been accompanied by rising dispersion of markups, indicating that changes in misallocation may be an important angle to consider. Barkai (2020) instead documents a rise in the profit share, while Karabarbounis and Neiman (2019) argue that, although the profit share has risen since the 1980s, it fell during the 1970s and is no higher today than it was in the 1960s. Autor et al. (2020) study U.S. Census data for the 1982–2012 sample for six large sectors and document rising concentration within four-digit industries, arguing that this trend mainly reflects reallocation towards large and productive firms. Gutiérrez and Philippon (2017) instead examine estimates of firm concentration at the economy-wide level and argue that it has risen over time, while Rossi-Hansberg, Sarte and Trachter (2021) document that "local market concentration" has declined.[3] Much of this literature has focused on the measurement of market power, while we study the impact of antitrust on these indicators and attempt to relate them to aggregate outcomes.

Closest to our efforts are Babina et al. (2023) and Besley et al. (2021). The latter study exploits an antitrust indicator based on the stringency of antitrust laws and policies to examine how antitrust affects firm profitability for a ten-year sample of a large number of firms in a cross-section of 94 countries. They test the hypothesis that antitrust stringency primarily affects firms in non-traded sectors and find evidence in favor of this hypothesis.[4] Babina et al. (2023), like us, study the U.S. and focus on cases pursued by the USDOJ during 1971–2018. They estimate the impact of antitrust at the state-industry level for non-traded sectors by comparing outcomes in states targeted by the DOJ with those not targeted. They find that antitrust enforcement actions stimulate payroll employment and wages and increase the labor share and business formation. Our study complements theirs in a number of dimensions. First, while Babina et al. (2023) focus on non-traded sectors, we

[3] Berry, Gaynor and Scott Morton (2019) warn against interpreting correlations between output, or other outcomes, and market concentration measures as causal evidence on the impact of market power since market concentration can change for many reasons other than market power.

[4] Buccirossi et al. (2013) similarly relate a proxy for competition policy to outcomes for a panel of 22 countries. They focus on productivity and find that stricter competition policy is associated with higher productivity growth.

focus on publicly listed firms, which enables us to estimate the impact of antitrust actions at the firm level and construct our antitrust indicators from these estimates. Second, we collect information on antitrust actions for a longer sample period and include both the FTC and the USDOJ. Third, we examine a broader set of antitrust violations. Fourth, we estimate U.S. outcomes at both the sector and aggregate levels. We see the two studies as complementary because they examine different aspects of antitrust.

2 U.S. Antitrust

2.1 Legislative Background

The main focus of U.S. antitrust legislation is to avoid harm to consumers from anti-competitive behavior by the corporate sector and to sustain high quality and low prices for goods and services sold on the market. Formal U.S. antitrust legislation dates back to the **Interstate Commerce Act of 1887**. This act was introduced to regulate the U.S. railway industry, which was considered to engage in unreasonable business practices resulting in excessively high prices due to a lack of competition and collusion.[5] Broader antitrust legislation followed swiftly in 1890, when the U.S. Congress passed the **Sherman Antitrust Act** to combat "business trusts." The next major legislative step in antitrust regulation was taken in 1914 with the passage of the **Clayton Antitrust Act of 1914** and the **Federal Trade Commission Act**. The Sherman and Clayton Acts, and later amendments, still provide the backbone of federal U.S. antitrust legislation, but their interpretation and implementation have evolved over time through the U.S. judicial system.[6]

The Sherman Act prohibits "*every contract, combination, in the form of trust or otherwise, or conspiracy, in restraint of trade or commerce among the several States, or with foreign nations*" (Section 1) and monopolization and attempts to monopolize (Section 2). Of course, not all contracts dealt with in Section 1 are seen as anti-competitive, but some acts are viewed as anticompetitive *per se* and are prohibited outright. Per se violations include:

[5] In practice, the legislation also had ramifications for non-railroad shipping; see, e.g., Gilligan et al. (1989) for an analysis.

[6] In addition, U.S. states typically have enacted their own antitrust laws targeted at anti-competitive behavior within their states, and firms are also subject to foreign entities' antitrust regulation when conducting business operations under those jurisdictions.

- *Price fixing*: Agreements between competitors regarding the prices of products or conditions offered;
- *Bid rigging*: Coordination of bidding behavior among competitors that undermines the bidding process;
- *Market or customer allocation*: Agreements among competitors not to compete for customers;
- *Boycotts*: Refusals to deal with certain customers or restrictions on the conditions under which they can be served;
- *Tying*: When firms with market power condition the availability of one product on the purchase of one or more other, separate goods.

Other actions are evaluated in a broader context under a "Rule of Reason" principle, which applies to *restraints on supply*, agreements in vertical relationships, and *exclusive dealing*, that is, exclusivity arrangements between suppliers and retailers. In either case, courts may deem certain actions to have sufficient pro-competitive benefits for the Rule of Reason to permit these practices.

Section 2 of the Sherman Act prohibits firms from monopolizing, attempting to monopolize, or conspiring to do so. Such actions may be undertaken unilaterally or jointly with other firms in a coordinated manner (as with Section 1 violations). While it is not illegal for a firm to be a monopoly as such, Section 2 regulates behaviors aimed at acquiring or maintaining such a position through unreasonable behavior. A *monopoly* is defined as a situation where a firm has "market power" in the sense of being able to control prices or the level of competition within a "market." The Sherman Act also contains a Section 3, but it does not define separate regulations and instead extends Sections 1 and 2 to the District of Columbia and U.S. territories.

The Sherman Act was left deliberately vague in many of its details so that U.S. courts could develop its implementation. This aspect of the act was eventually seen as a weakness, and antitrust reform became a topic of the 1912 U.S. presidential election. After the election of President Woodrow Wilson, Congress passed the Federal Trade Commission Act and the Clayton Antitrust Act in 1914. These acts were meant to make the legislation more specific and forward-looking and to improve antitrust enforcement. The Clayton Antitrust Act prohibits specific actions when they *may* lessen competition. Actions include:

- *Price discrimination* between different buyers when it substantially lessens competition or tends to create a monopoly (Section 2);
- *Tying and exclusivity* of sales when these substantially lessen competition (Section 3);
- *Mergers and acquisitions* that may either significantly lessen competition or create a monopoly (Section 7);
- *Forming interlocking boards* on competing companies when a merger between them would violate antitrust law (Section 8).

In an amendment to the Clayton Act that extends Section 2, the **Robinson-Patman Act of 1936** further bans a set of price discriminatory actions. Section 7 of the Clayton Act was expanded in the **1950 Celler-Kefauver Amendment**. Before the amendment, Section 7 specifically prohibited stock acquisitions likely to lessen competition but did not apply to asset acquisitions or, somewhat surprisingly, to actual mergers. The Celler-Kefauver Amendment extended the legislation to asset purchases and mergers.

In an important later addition to the formal antitrust legislation, the **Hart-Scott-Rodino Antitrust Improvements Act of 1976** introduced a number of amendments to U.S. antitrust legislation. Principally, the act requires firms to pay a filing fee and to pre-notify and register mergers and acquisitions that affect U.S. commerce and exceed certain limits on the size of the transactions, assets, or net sales of the parties involved. Companies involved in such transactions must subsequently await approval from the antitrust authorities before they can implement any integration of their business activities. Failure to pre-notify or completion of the transaction prior to approval can trigger a civil penalty.

2.2 Enforcement

Antitrust enforcement was initially under the remit of the U.S. Attorneys and the Attorney General. Today, federal antitrust legislation in the U.S. is enforced by the **Antitrust Division** (the ATR) of the U.S. Department of Justice (the USDOJ) and by the **Federal Trade Commission (FTC)**. At the state level, antitrust is enforced by state attorneys general, and private parties can also seek enforcement against competitors. Our analysis focuses on the enforcement of federal antitrust laws by the FTC and the USDOJ.

The USDOJ was involved in antitrust enforcement from the passage of the Sherman Act, but with scarce resources and without a dedicated antitrust division, its role was limited. In 1919, Attorney General A. Mitchell Palmer initiated a reorganization of the USDOJ that led to the formation of the ATR as a dedicated antitrust enforcement team; see Werden (2018). The mission of the ATR is to "*promote economic competition through enforcing and providing guidance on antitrust laws and principles*."

The ATR can prosecute individuals and firms deemed in violation of U.S. antitrust laws by filing criminal lawsuits, and it can instigate civil actions. In pursuing violators of antitrust laws and regulations, it can ask courts to prohibit future violations and require offenders to remedy past violations. For certain violations, offenders may face large fines and, in the case of individuals, up to 10 years' imprisonment. The choice of whether to pursue violations as criminal or civil cases depends on the prosecutor's views about the seriousness of the alleged offense and the strength of the evidence. Furthermore, only certain offenses are viewed as criminal, and some have only recently become felonies (cartelization, for example, became a felony in 1974). In many cases, there is therefore a somewhat fluid distinction between civil non-merger cases and criminal cases. The ATR also functions as an advocate for competition and provides guidance on antitrust laws to firms through business reviews. The ATR has permission to fill more than a thousand positions and currently employs more than 50 PhD economists and fewer than 400 attorneys.

The FTC was created in 1914 with the passage of the Federal Trade Commission Act. It is an independent agency of the U.S. government and enforces noncriminal antitrust and consumer-protection laws. Its mission is to "*prevent business practices that are anticompetitive or deceptive or unfair to consumers*." The FTC investigates competition issues brought to its attention by reports from consumers or firms, pre-merger notification filings, congressional inquiries, or reports in the media. It can seek voluntary compliance, file administrative complaints, and initiate federal litigation. The FTC has more than a thousand staff, including around 300 lawyers, and it employs approximately 75 PhD economists.

Jointly, the ATR and the FTC therefore possess significant analytical and technical expertise in antitrust, although their joint technical staffing and budgets are not large compared with, for example, the Federal Reserve System.[7]

The USDOJ enforces criminal antitrust cases related to Sections 1 and 2 of the Sherman Act, and it can also initiate civil actions requesting injunctive relief and damages. In cases of litigation, the USDOJ also enforces Sections 2 and 3 of the Clayton Act, but the FTC takes the lead in administrative proceedings. Sections 7 and 8 of the Clayton Act are jointly enforced by the FTC and the USDOJ. The FTC may also enforce violations of Sections 1 and 2 of the Sherman Act, but not in criminal cases. The HSR Act of 1976 is also jointly enforced, with both agencies engaging in pre-consummation reviews and requests for further details. For such cases, the agencies coordinate to determine which agency carries out the review and pursues potential violations. The FTC may also investigate cases under Section 5 of the Federal Trade Commission Act, which deals with unfair methods of competition and adds to the USDOJ's competencies under the Sherman Act.

2.3 Sources and Methodology

We collect data on the universe of federal antitrust investigations carried out by the FTC and the ATR for sample periods that cover 1954–2023 and 1957–2023, respectively. The primary source of information about antitrust investigations is the Commerce Clearing House Trade Regulation Reporter (the CCH Trade Reporter), also referred to as the CCH Bluebook. Whenever possible, we double-check the information from the CCH Trade Reporter with antitrust case documents published on the websites of the USDOJ and the FTC. In certain cases, we also use information from Forms 10-K, 8-K, and 10-Q filed with the SEC.

Following Posner (1970) and Gallo et al. (2000), we make several adjustments to the raw CCH Trade Reporter data. First, the CCH Trade Regulation Reporter frequently contains several updates on the same investigation as it proceeds through different legal stages. Counting such updates as separate cases would lead to substantial double-counting. In some cases, though, further defendants are added during the course of a case. To deal

[7] In comparison, the Federal Reserve Board employs more than 400 PhD economists.

with these issues, unique cases are counted from the dates on which they were initially filed, but if further defendants are added during the process, we record the investigation date for these defendants as the later date. Second, it is not uncommon for one case to trigger further separate cases. This happens, in particular, in cases where employees are investigated for criminal offenses related to investigations of corporations. We choose not to control for such possible double-counting.

For each antitrust case filed by the ATR or the FTC, we record the following information:

- the docket number;
- the type of defendant (firm or individual);
- the case type: civil non-merger, criminal or merger;
- the alleged violation(s) of antitrust legislation;
- the court where the case was filed;
- the names of firms and/or individuals involved;
- the dates of significant case developments.

For cases that involve firms rather than individuals ("natural persons"), we check whether the companies involved were publicly traded at the time of the case opening and, if so, link them to their stock market tickers and to Compustat accounting data. This mapping of firms named in antitrust cases to publicly traded firms is very labor-intensive because firms are often referred to using slightly different names and because of typos in antitrust filings.

2.4 Broad Trends

Table 1 contains summary statistics for the history of antitrust cases. In the sample period that we examine, the antitrust authorities initiated a total of 6,387 investigations, with the USDOJ accounting for approximately two-thirds of them. Panel A of Figure 1 illustrates the total number of federal antitrust cases over the 1954–2023 sample for the FTC and the 1957–2023 sample for the ATR separately, as well as the sum of the two. As is evident, the late 1950s and early 1960s were a period of very high antitrust activity relative to the rest of the sample. Apart from this early period, the total number of cases has been rather stable over time, with two exceptions. First, the USDOJ was very active from the mid-1970s to the mid-1980s. Second, there appears to have been some decline in antitrust activity at the very end of the sample.

Table 1: Antitrust Investigations: Summary Statistics

		Defendants				
				Firms		
	Cases	**Total**	**Individuals**	**Total**	**Non-Listed**	**Listed**
Total	6,387	15,829	4,261	11,568	8,564	3,004
FTC	2,266	3,946	635	3,311	2,284	1,027
DOJ	4,121	11,883	3,626	8,257	6,280	1,977
Civil non-merger	2,267	5,582	936	4,646	3,670	976
Criminal	2,577	7,250	3,205	4,045	3,407	638
Mergers and acquisitions	1,524	2,935	108	2,827	1,451	1,376
Mixed	19	62	12	50	36	14

Notes: The FTC sample covers 1954–2023, and the DOJ sample covers 1957–2023. The first column reports the number of antitrust cases initiated by the antitrust authorities. The remaining columns report the number of defendants named in those cases. "Listed firms" refers to firms that were listed on a U.S. or foreign stock exchange when the case was filed. Source: authors' calculations based on the CCH Trade Reporter and information published by the FTC and DOJ on their websites.

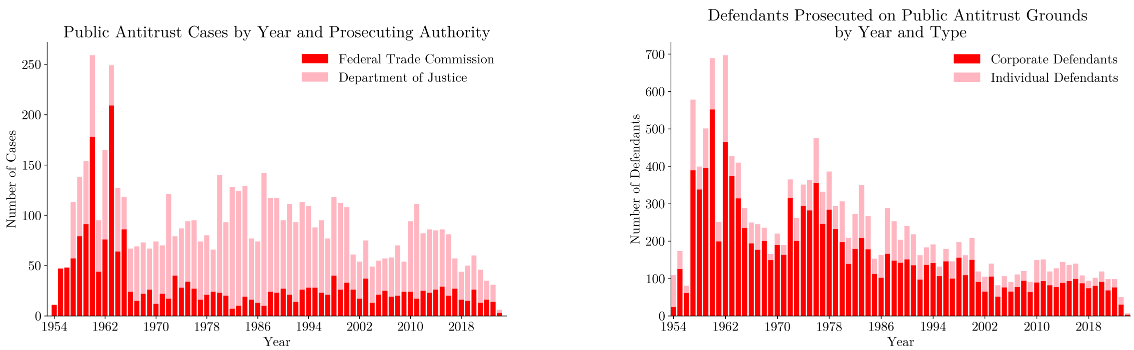
Antitrust cases often involve multiple defendants, and defendants may be either firms or individuals ("natural persons"). Furthermore, firms may be either publicly traded ("listed") or not. In total, the 6,387 antitrust cases in our sample involve 15,829 defendants, the majority of which are corporations (11,568 in total). Among corporations, publicly traded firms account for approximately 26 percent. Thus, publicly traded firms, which tend to dominate the right tail of the firm-size distribution, are significantly overrepresented given that they account for only a tiny fraction of the total number of businesses in the U.S.

Panel B illustrates the trends in the total number of firms or individuals indicted for antitrust violations. As is evident, there is considerable time variation in the number of corporate firms investigated and, in contrast to the number of cases, a marked negative trend over time in the number of firms indicted. During 1957–1964, an average of 379 firms were involved in antitrust investigations each year; this number declines to 238 firms per year for the 1965–1980 sample and further to 81 firms per year post-2000. The number of individuals investigated for antitrust violations instead remains relatively stable over the sample, apart from peaking at 116 persons during 1957–1964.

Figure 1: Antitrust Trends

Panel A: By Authority

Panel B: By Defendant Type



Another important consideration concerns the type of antitrust violation involved. Of the 6,387 antitrust cases in the sample, 2,267 relate to civil non-merger antitrust violations, 2,577 are criminal cases (which can only be pursued by the USDOJ), 1,524 relate to civil merger violations, and a small number (19) are mixed. Focusing on corporate antitrust defendants, civil non-merger violations account for approximately 40 percent of the cases, criminal cases for 35 percent, and M&A for the remaining 25 percent. For listed firms, civil merger cases instead dominate the indictments and account for 46 percent of the defendants (1,376 defendants), while civil non-merger violations account for 32 percent and criminal cases for the remaining 21 percent.

Panel A: All Cases

Panel B: Firms Prosecuted

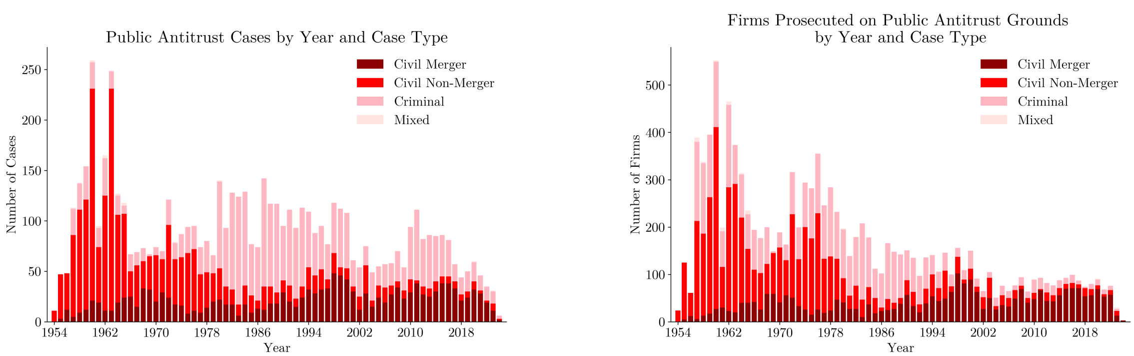


Figure 2: Antitrust Trends by Case Type

Panel A of Figure 2 illustrates the development in the number of antitrust *cases* over time across the four broad categories of antitrust violations, while Panel B illustrates the same time series for the number of *firms* indicted. This figure demonstrates a remarkable change in antitrust focus in the U.S. over the 70-year period that we examine. From 1954 to the late 1970s or early 1980s, the antitrust authorities frequently initiated investigations related to civil non-merger violations, but from the early 1980s, such civil-non-merger-related antitrust cases became rare. During 1957–1980, civil non-merger-type violations accounted for an average of 63 percent of all cases, while the corresponding share from 1981 to the end of the sample is only 14 percent. The decline in the relative frequency of civil non-merger cases is accounted for by increases in the importance of both civil merger cases, which accounted for 16 percent of all cases during 1957–1980 but 30 percent of all cases post-1980, and criminal cases, whose share increased from 21 percent to 56 percent across those two subsamples. As mentioned earlier, whether an offense is deemed criminal depends both on the type of offense and on an evaluation of the seriousness of the alleged offense and the robustness of the evidence. Thus, the decline in the importance of civil non-merger-type cases appears to be related to a combination of the growing importance of civil merger cases and a shift from civil non-merger cases to criminal cases. Consistent with this, Ghosal (2011) tests for a structural break in antitrust enforcement during 1958–2002 and concludes that there was a structural break in 1979, with a shift towards criminal cases. New to our evidence is the significant growth in the importance of civil-merger-related cases.

The corresponding trends at the firm level shown in Panel B illustrate a similar decline in the importance of civil non-merger violations, which accounted for 53 percent of all firms indicted for antitrust violations up to 1980 but only 19 percent in the post-1980 sample. An important difference relative to Panel A is that, as far as the number of firms pursued for antitrust violations is concerned, the importance of criminal cases is also declining: these cases accounted for 34 percent of the total number of firms indicted during 1957–1980 but only 19 percent since 2000. Thus, civil merger cases have become the main reason for antitrust cases raised against firms.

In Figure 3, we focus on antitrust cases raised against *publicly listed companies*. For these large firms, the structural change in the composition of the antitrust violations for which firms are prosecuted is both very visible and very significant. In the 1954–1980 sample, M&A-related violations accounted for 22 percent of the total number of firms indicted for antitrust violations, while in the 1981–2023 sample, the corresponding share is 79 percent. Thus, the changing focus of antitrust discussed above is particularly evident for public firms. Furthermore, for this subset of defendants, the frequency of antitrust case openings also displays a negative trend over the sample, occurring around the same time as the shift in the focus of the antitrust authorities towards M&A-related violations. In particular, the number of publicly listed firms prosecuted for antitrust violations declines from around 71 per year in the 1954–1980 sample to 30 firms per year post-1980. This decline appears to be mainly the product of fewer investigations of publicly listed firms initiated by the USDOJ, while the number of firms investigated by the FTC remains roughly unchanged.

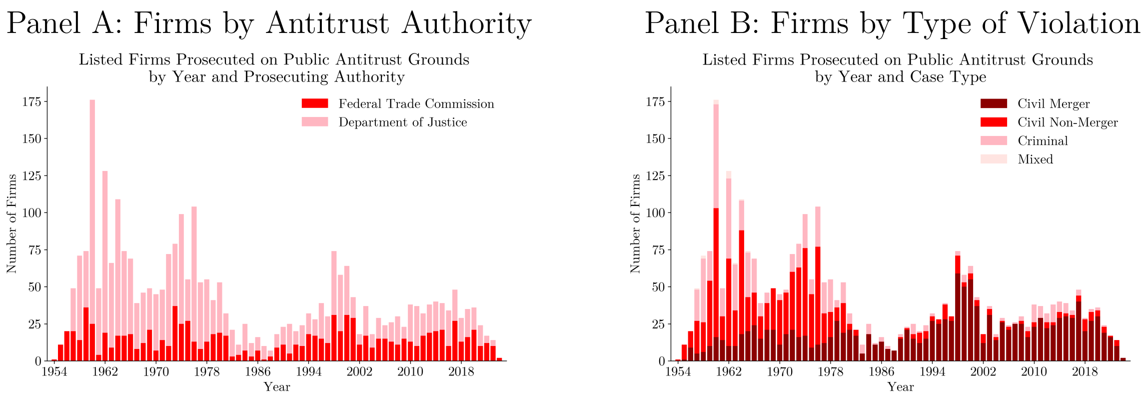


Figure 3: Publicly Listed Firms

An important question is the extent to which firms prosecuted for antitrust violations differ in observable ways from other firms. In Table 2, we report several statistics comparing publicly listed firms prosecuted for antitrust violations with their competitors. We define competitors as publicly listed firms that operate within the same three-digit NAICS sector as the prosecuted firm. By construction, publicly listed firms are large relative to all firms in the economy, but prosecuted firms also tend to be large relative to other publicly listed firms. We find that the prosecuted firms are larger than their competitors, regardless of whether size is measured by market value, operating income, capital expenditure, R&D expenditure, number of employees, or market share. On the other hand, in terms of sales per employee or sales relative to fixed assets, the prosecuted firms appear to be less productive.[8]

Table 2: Comparison of Means of Prosecuted Firms and Their Competitors

	Prosecuted	**Competitors**	**Difference**	***t*-statistic**	***p*-value**
Equity value	18,863.5	4,542.8	14,320.7	3.729	< 0.001
Operating income	1,929.3	405.1	1,524.2	6.899	< 0.001
Capital expenditure	874.5	148.4	726.2	7.147	< 0.001
R&D expenditure	844.7	189.1	655.6	2.708	0.007
No. of employees	45,196	7,746	37,449	10.483	< 0.001
Market share	0.210	0.018	0.193	23.936	< 0.001
Sales per employee	258.6	392.2	−133.6	−2.423	0.015
Sales/fixed assets	5.176	14.884	−9.708	−16.509	< 0.001

Notes: Equity value, operating income, capital expenditure, and R&D expenditure are measured in millions of 2017 dollars using the GDP deflator. Sales per employee is measured in thousands of 2017 dollars. Equity value is the market value of the firm's equity. Operating income is measured before depreciation. Market share is computed at the three-digit NAICS level. All entries are means.

Figure 4 illustrates the prosecution rate computed over the entire sample for publicly listed firms across deciles of the firm-size distribution, as measured by either employment or three-digit NAICS market shares. There is a very strong size dependence in the prosecution rates: firms in the top decile of the employee distribution (market-share distribution) have a 17 percent (24 percent) frequency of having been prosecuted at least once in the sample that we examine. By contrast, firms in the bottom half of the size distributions have a near-zero chance of being observed in the sample. It is important to stress that these numbers refer to the size distributions of publicly listed firms, which are already right-skewed. Thus, as far as the corporate sector is concerned, antitrust indictments are heavily concentrated among large firms.

3 The Impact of Antitrust Investigations on Firm Valuations

We now turn to estimating the impact of antitrust indictments on firm valuations. The fact that a firm is prosecuted for violations of antitrust legislation is, of course, not a random event: to the extent that the allegations are correct, such firms are likely to have earned abnormal profits prior to the case opening date. Therefore, difference-in-differences-type estimators cannot be applied. Moreover, when prosecutions lead to changes in business practices,

[8] Using production-function-based measures of TFPR, prosecuted firms are instead more productive than their competitors.

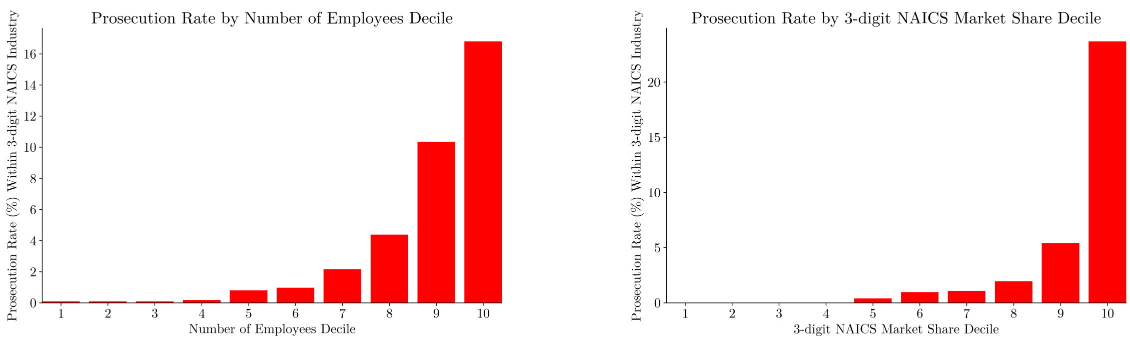


Figure 4: Prosecution Rates

antitrust can affect both the firms pursued for antitrust violations *and* their competitors, implying that the SUTVA (Stable Unit Treatment Value Assumption) condition may be invalid, which rules out the use of cross-sectional estimators unless one is willing to explicitly model spillovers between firms.

For that reason, we adopt an event-study approach. In particular, focusing on publicly listed firms, we exploit information on equity returns to evaluate the impact of antitrust prosecutions on firm valuations by comparing realized equity returns after a firm has been indicted for violations of antitrust legislation with the predicted returns in the absence of such an indictment. For the latter, we use a parametric model of equity returns estimated on data prior to the indictment. We interpret these estimates as measuring how the expected loss of profits derived from anticompetitive practices affects prosecuted firms' valuations. The underlying idea is that (i) firms engage in anticompetitive practices to increase profit margins and (ii), when a firm is prosecuted for such practices, market participants expect some of them to end, thereby reducing expected profits and the market value of the firm's equity.

3.1 Methodology

There are three steps to the procedure. In the first step, for each listed firm indicted for antitrust violations, we estimate the determinants of equity returns on a pre-treatment sample ahead of the case opening date. We apply a simple three-factor Fama-French model and estimate the loadings on the return determinants using daily equity return data for a 12-month period starting 13 months before the case opening date and ending one month prior to this date. We leave a one-month gap between the treatment (the case opening) and the estimation sample to eliminate concerns about contamination from the subsequent antitrust case opening. In the second step, we compute daily excess and cumulated daily excess returns on the treated firms' equity in windows around the case opening date. Excess returns are defined as the differences between realized returns at the firm level and those implied by the pre-treatment estimates of the loadings on the factors determining the firms' equity returns. Third, we convert the cumulated excess returns into dollar equivalents for each case by combining them with firm market capitalizations at the time of the antitrust case opening.

We include all firms pursued for antitrust violations that, at the time of the antitrust case opening, were listed on either U.S. or foreign stock exchanges.[9] For companies listed in the U.S., we obtain daily equity prices from the Center for Research in Security Prices (CRSP), and for companies listed on foreign stock exchanges, we use equity price data from LSEG (formerly Refinitiv).

We date the antitrust cases by the case opening dates. Posner (1970) and Gallo et al. (2000) show that the success rates of the USDOJ have been very high since the 1950s, so we work under the assumption that the equity market assumes that cases will lead to antitrust intervention in commercial activities. Let $t^a_{i,j}$ denote the day of the announcement of antitrust investigation i against firm j, let $r_{j,t}$ denote the equity return of firm j at date t, and let r^f_t denote the risk-free rate. The first stage then consists of least squares estimation of the Fama–French three-factor model:

$$r_{j,t} - r^f_t = \alpha_{i,j} X_t + \nu_{j,t}, \qquad t \in [t^a_{i,j} - T_1, t^a_{i,j} - T_2] \tag{1}$$

where $X_t = (1, smb_t, hml_t, r_{m,t})$ denotes a constant and the three standard Fama-French factors: smb_t is the "small minus big" factor meant to capture a size premium, hml_t is the "high minus low" factor capturing a value premium, and $r_{m,t}$ is the equity market excess

[9] We have some attrition among firms listed outside the U.S. when they have changed ownership after the antitrust investigation.

return measured as the return on the market equity portfolio less the risk-free rate.[10] $T_1 = 13$ months and $T_2 = 1$ month define the window in which the factor loadings are estimated.

In the second stage, we calculate excess equity returns for each prosecuted firm in windows around the case opening date. The excess returns at horizon s are defined as the difference between realized returns and those predicted on the basis of the factor loadings estimated in equation (1):

$$R_{i,j,s} = \left(r_{j,t^a_{i,j}+s} - r^f_{t^a_{i,j}+s} \right) - \widehat{\alpha}_{i,j} X_{t^a_{i,j}+s}, \qquad s \in [s_1, s_2] \tag{2}$$

where $\widehat{\alpha}_{i,j}$ denotes the least squares estimate of $\alpha_{i,j}$ and $[s_1, s_2]$ denotes the event window that we evaluate. We then use these estimates to compute cumulated excess returns:

$$AR_{i,j,s} = \sum_{h=s_1}^{s} R_{i,j,h} \tag{3}$$

Finally, we convert the cumulated excess returns into dollar values by multiplying them by firm j's market capitalization at the case-opening date $t^a_{i,j}$:

$$Q^M_{i,j,s} = AR_{i,j,s} \times cap_{j,t^a_{i,j}} \tag{4}$$

where $cap_{j,t^a_{i,j}}$ is the market capitalization of firm j at date $t^a_{i,j}$.

3.2 Results

It is important for our analysis that the first-stage regressions of daily returns based on estimates from the Fama-French three-factor model have at least some explanatory power. Across all treatments (2,707 cases), the mean and median R^2 statistics are 20.9 percent and 17.2 percent, respectively, which we find satisfactory. Figure 5 illustrates the distribution of R^2 statistics, indicating that the fit is good apart from the bottom 20 percent of cases, where the R^2 is below 7.5 percent. Given the noisy nature of daily returns, we believe that these results are encouraging.

[10]Alternatively, one might use a Fama-French 5-factor specification, but data for the two additional factors are not available for the entire sample period that we study. For the sample period that allows the use of the 5-factor model, results are very similar to the 3-factor model.

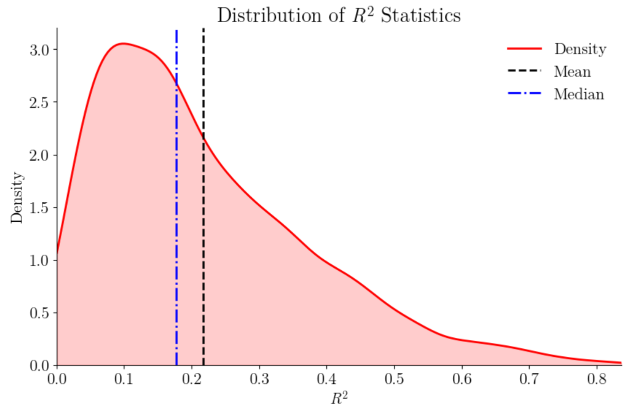


Figure 5: Distribution of R^2 Statistics Associated with Equation 1

We report average firm-level estimates of excess and cumulative excess returns, $R_{i,j,s}$ and $AR_{i,j,s}$, for each horizon (measured in trading days):

$$\overline{R}_{z,s} = \frac{1}{N_{z,s}} \sum_{(i,j)\in V_{z,s}} R_{i,j,s} \tag{5}$$

$$\overline{AR}_{z,s} = \frac{1}{N_{z,s}} \sum_{(i,j)\in V_{z,s}} AR_{i,j,s} \tag{6}$$

where $V_{z,s}$ denotes the set of usable firm–case observations in selection z at horizon s, and $N_{z,s} = |V_{z,s}|$ is the corresponding sample size.

Tables 3 and 4 contain the estimates of the average excess returns and average cumulated excess returns, respectively. We report estimates for all cases and separately for civil non-merger and criminal cases and for civil merger cases. The excess return estimates are reported for horizons ranging from five trading days before the case opening date to 60 trading days after it. For the estimates of $\overline{AR}_{z,s}$, for reasons that will become clear below, we report results cumulating excess returns from two days before the case opening day. Care must be taken when conducting inference in this environment because many investigations involve multiple firms, creating cross-sectional correlation in excess and cumulated excess returns. Moreover, equity returns may be subject to high event-day volatility due to market uncertainty about the impact of antitrust cases. To take these issues into account, we report p-values for the null hypothesis that the (cumulated) excess returns equal zero using the test statistic proposed by Kolari and Pynnönen (2010). This test statistic modifies the test statistic of Boehmer, Musumeci and Poulsen (1991), developed to address volatility, to additionally account for cross-sectional correlation in abnormal (cumulated) returns.[11]

Table 3: The Impact of Antitrust Case Filings on Excess Returns

	All Cases			**Civil Non-Merger and Criminal**			**Civil Merger**		
s	N	$\overline{R}_s$	p	N	$\overline{R}_s$	p	N	$\overline{R}_s$	p
−5	2,707	0.00	0.45	1,262	0.02	1.00	1,430	−0.01	0.28
−4	2,707	−0.01	0.77	1,262	−0.01	0.69	1,430	−0.01	0.97
−3	2,707	0.03	0.73	1,262	−0.01	0.36	1,430	0.05	0.77
−2	2,707	−0.11	< 0.01	1,262	−0.06	0.03	1,430	−0.16	0.02
−1	2,707	−0.03	0.70	1,262	−0.05	0.46	1,430	−0.02	0.93
0	2,707	−0.44	< 0.01	1,262	−0.45	< 0.01	1,430	−0.42	< 0.01
1	2,686	−0.30	< 0.01	1,262	−0.27	< 0.01	1,409	−0.31	< 0.01
2	2,665	0.02	0.83	1,262	−0.02	0.43	1,388	0.05	0.77
3	2,650	−0.02	0.49	1,262	0.01	0.35	1,373	−0.04	0.94
4	2,643	−0.04	0.04	1,262	0.01	0.46	1,366	−0.08	0.04
5	2,636	−0.02	0.73	1,262	0.02	0.42	1,359	−0.04	0.27
10	2,606	−0.03	0.46	1,260	0.04	0.59	1,331	−0.11	0.13
20	2,583	−0.06	0.19	1,259	−0.11	0.06	1,310	0.00	0.84
30	2,569	0.01	0.90	1,259	0.03	0.70	1,296	−0.02	0.91
40	2,556	−0.01	0.58	1,259	−0.04	0.14	1,283	0.03	0.51
50	2,544	0.06	0.08	1,259	0.06	0.09	1,271	0.05	0.39
60	2,532	0.01	0.61	1,256	−0.02	0.27	1,262	0.04	0.91

Notes: s denotes the number of trading days relative to case opening, and N is the number of usable firm–case observations at that horizon. $\overline{R}_{z,s}$ is the average excess return at horizon s, as defined in Equation (5). The p-value is associated with the Kolari and Pynnönen (2010) test of the null hypothesis that the average excess return is zero.

We find a spike in negative excess returns on the case opening day for all cases and for civil non-merger and criminal cases and merger cases separately. The negative excess returns persist on the day after the case opening, which could be due either to the timing of the announcement of the antitrust case filing within a trading day or to short-run persistence in the updating of investors' return beliefs. For both trading days, we can firmly reject the null that the average excess returns are zero for all cases, civil non-merger and criminal cases, and mergers separately.[12] After the first two trading days, excess returns are close to zero and

[11]We computed a battery of other test statistics, including the Kolari and Pynnönen (2011) nonparametric test statistic, which, in addition to accounting for cross-sectional correlation and event-day volatility, is robust to serial correlation and nonnormality. In terms of the significance of the estimated returns, most results are identical.

[12]The nonparametric test statistic of Kolari and Pynnönen (2011) also rejects the null for all cases and for civil non-merger and criminal cases.

insignificantly different from zero, with very few exceptions. Prior to the case opening date, we find significant negative excess returns two days ahead of the case filing for civil mergers (and for all cases), while the evidence is slightly less clear for civil non-merger and criminal cases. It is unclear why excess returns should be negative prior to the case opening date for these cases. Although the estimated abnormal returns at this horizon are much smaller than those on the case opening date and the following trading day, we can reject the null hypothesis that they are zero.

Table 4: The Impact of Antitrust Case Filings on Cumulative Excess Returns

	All Cases			**Civil Non-Merger and Criminal**			**Civil Merger**		
$[s_1, s_2]$	N	$\overline{AR}_{s_2}$	p	N	$\overline{AR}_{s_2}$	p	N	$\overline{AR}_{s_2}$	p
$[-2,-1]$	2,707	-0.14	0.01	1,262	-0.11	0.04	1,430	-0.17	0.12
$[-2,0]$	2,707	-0.58	< 0.01	1,262	-0.56	< 0.01	1,430	-0.59	< 0.01
$[-2,1]$	2,686	-0.90	< 0.01	1,262	-0.84	< 0.01	1,409	-0.94	< 0.01
$[-2,2]$	2,665	-0.90	< 0.01	1,262	-0.86	< 0.01	1,388	-0.92	< 0.01
$[-2,3]$	2,650	-0.94	< 0.01	1,262	-0.86	< 0.01	1,373	-1.01	< 0.01
$[-2,4]$	2,643	-0.99	< 0.01	1,262	-0.84	< 0.01	1,366	-1.11	< 0.01
$[-2,5]$	2,636	-1.01	< 0.01	1,262	-0.82	< 0.01	1,359	-1.16	< 0.01
$[-2,10]$	2,606	-1.18	< 0.01	1,260	-0.82	< 0.01	1,331	-1.51	< 0.01
$[-2,20]$	2,583	-1.15	< 0.01	1,259	-0.69	< 0.01	1,310	-1.60	< 0.01
$[-2,30]$	2,569	-1.41	< 0.01	1,259	-1.04	< 0.01	1,296	-1.80	< 0.01
$[-2,40]$	2,556	-1.38	< 0.01	1,259	-0.77	0.01	1,283	-2.02	< 0.01
$[-2,50]$	2,544	-1.38	< 0.01	1,259	-0.64	0.01	1,271	-2.12	< 0.01
$[-2,60]$	2,532	-1.60	< 0.01	1,256	-1.04	< 0.01	1,262	-2.17	< 0.01

Notes: $[s_1, s_2]$ denotes the inclusive event window, and N is the number of usable firm–case observations at its final horizon. $\overline{AR}_{z,s}$ is the average cumulative excess return from trading day -2 through trading day s, relative to case opening, as defined in Equation (6). The p-value is associated with the Kolari and Pynnönen (2010) test of the null hypothesis that the average cumulative excess return is zero.

Given the evidence on abnormal returns two days ahead of the case opening, Table 4 reports average cumulated excess returns from this trading day. We find large negative cumulated excess returns and reject the null hypothesis that they equal zero at very high significance levels at all horizons from two days prior to the case opening up to 60 trading days after the case opening. Quantitatively, the impact on average cumulated excess returns is very similar for the two types of antitrust violations at short horizons going up to 2–3 trading days after the case opening. After this, the average cumulated excess returns stabilize for civil non-merger and criminal cases at a level corresponding to around -0.9 to -1 percent, while they rise gradually with the forecast horizon for mergers until 50 trading days after the case opening date, and stabilize at a level closer to -2 percent. Bosch and Eckard (1991) estimate average excess return and average cumulated excess returns for 127 firms indicted for price fixing under Section 1 of the Sherman Act in the 1962–1980 period. They estimate a spike in negative abnormal returns on the indictment date of -0.75 percent and of -0.33 percent the day before, with the former (latter) being significant at the 99 percent (90 percent) level. These estimates are larger than those that we estimate for civil non-merger and criminal cases, which could be due to their selection relating to more severe violations. However, in terms of cumulated excess returns, the point estimates of the abnormal returns in their study and in ours are remarkably similar: Bosch and Eckard (1991) estimate that the cumulated excess returns from 5 trading days prior to the indictment to 5 days after are -0.78 percent, while the estimate over this window is -0.81 percent for the civil non-merger and criminal cases in our sample. Our study, however, includes an order of magnitude more firms and covers a much longer sample period.

As discussed above, we find a remarkable shift in antitrust focus around 1980, after which civil non-merger cases became very rare for publicly listed firms, while merger cases, if anything, became more common. For that reason, in Figure 6, we show the estimated average cumulated excess returns for the two types of antitrust cases, comparing the pre-1980 sample with a sample that starts in 1980. We show the average cumulated excess returns together with two-standard-error bands, which should be interpreted with care due to cross-sectional dependence and volatility. As is clear, average cumulated excess returns are significantly negative in the pre-1980 sample for both types of antitrust violations. In the post-1980 sample, we find significant negative cumulated excess returns for civil non-merger and criminal cases only for a few trading days after the case opening, after which they hover around zero.[13] For merger cases, the estimates for the post-1980 sample indicate significant negative cumulated excess returns, although they take longer to cumulate and stabilize at a lower level than in the pre-1980 sample.

One might speculate about the underlying reasons for this structural change. One

[13] A formal test of the null hypothesis that cumulated excess returns are zero that takes cross-sectional correlation and volatility into account fails to reject this hypothesis beyond four trading days. Taking serial correlation into account as well, the cumulated excess returns are significant for five trading days but only at the 10 percent level.

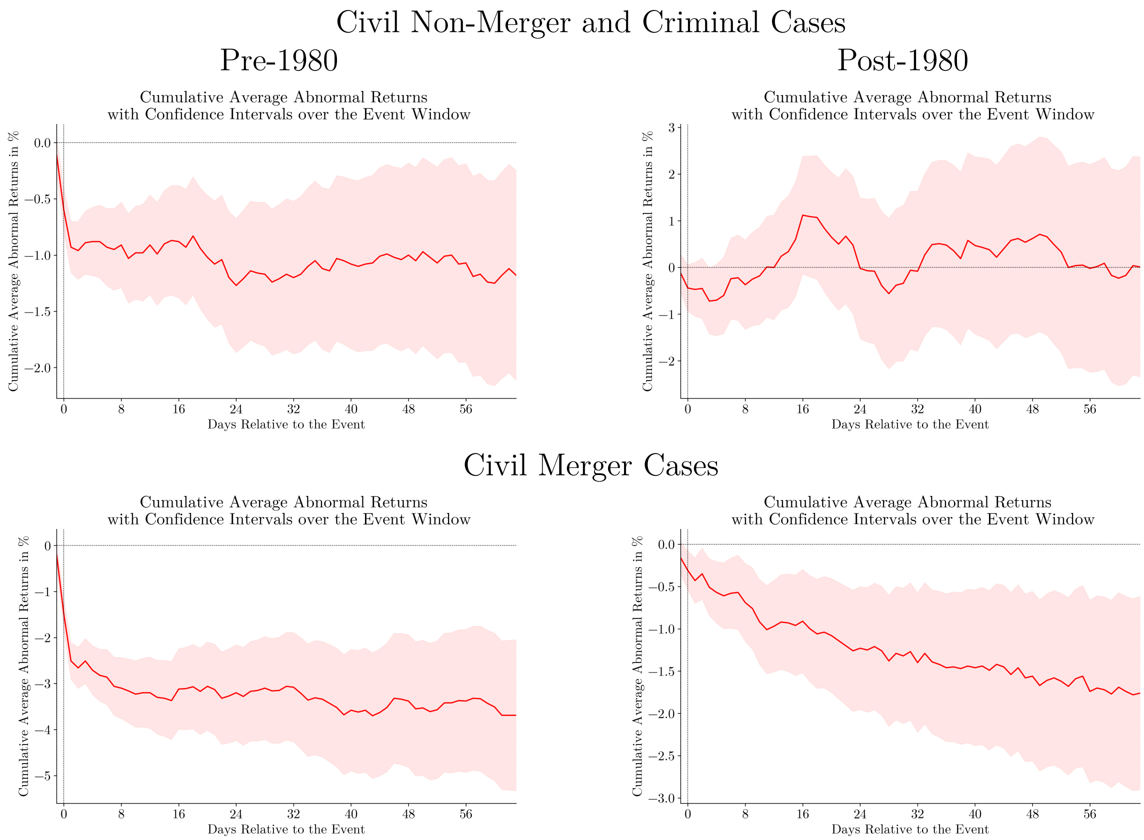


Figure 6: Average Cumulated Excess Returns on Prosecuted Firms' Daily Equity Returns Pre- and Post-1980

hypothesis is that it resulted from the introduction of the HSR Act of 1976, which meant that the antitrust authorities had to redirect a significant amount of their resources to pre-consummation reviews of large merger cases. Another possibility is that it reflects a change in antitrust orientation. In this respect, the structural change coincides with the publication of Bork (1978), who argued that antitrust should focus on how market power affects allocative efficiency rather than on large-scale business power (see, e.g., Crane (2014)). Robert Bork went on to state that "Congress designed the Sherman Act as a 'consumer welfare prescription,' " a statement that was subsequently often quoted in federal antitrust cases, indicating the potential impact of Bork's arguments. Third, starting in 1976 and perhaps related to Bork's arguments, the Manne Economics Institute for Federal Judges provided intensive economic training for federal judges that Ash et al. (2026) argue had a deep impact on their rulings. Lancieri et al. (2023) argue that weaker enforcement has occurred due to the influence of big business, which supported light enforcement in its own interests. These explanations clearly deserve attention in future research.

The average cumulated excess returns obscure differences across cases that are inevitable given the heterogeneity in both the seriousness of the antitrust cases and the volatility in daily equity returns. At the 21-trading-day horizon, the mean is significant for both types of antitrust cases before 1980, and the distribution of cumulated excess returns is skewed to the left. For mergers, the post-1980 distribution is relatively similar to the pre-1980 distribution, while for civil non-merger and criminal cases, the post-1980 distribution is approximately symmetric around zero and has a notably higher cross-case variance.

Given the differences in responsibility for antitrust enforcement between the FTC and the USDOJ, it is also interesting to examine whether there are significant differences in the impact of investigations by these two authorities on excess returns. We find that the point estimates of the average cumulated excess returns are larger for USDOJ cases than for FTC cases at all horizons that we estimate. For the USDOJ, the average cumulated excess returns stabilize at around −1.6 percent after 20 trading days, while for FTC cases, they stabilize at around −1 percent (also after 20 days). Taking sampling uncertainty into account, the differences are statistically significant for at least the first 8–10 trading days. This difference in the impact of antitrust investigations may arise because the USDOJ is responsible for more serious criminal cases, while the FTC can only raise cases associated with civil litigation.

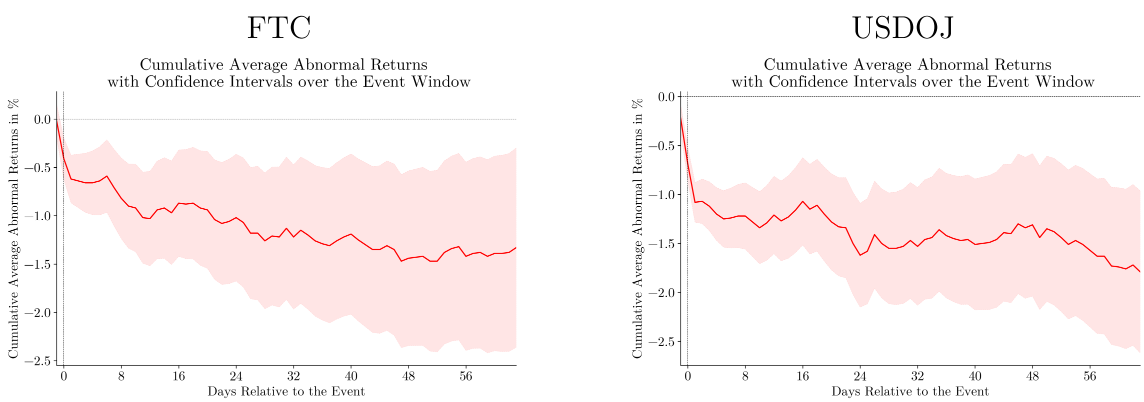


Figure 7: Average Cumulated Excess Returns: FTC vs USDOJ

Using the firm-level estimates of cumulated excess returns for prosecuted firms, we derive an indicator of antitrust activity by converting these estimates into dollar amounts as in equation (4). We evaluate the excess returns at 10 trading days and then sum over firms at each date. To minimize noise, we weight each firm-level estimate by the inverse variance of the excess returns. In practice, this reweights the measure towards units for which the event-study regressions are estimated with greater relative precision. Many event-study test statistics, such as Patell (1976), apply the same adjustment because it improves the statistical power of the tests. Given the high volatility of equity returns, we eliminate observations for which the estimated excess returns are positive after five trading days to minimize concerns about contamination by non-antitrust-related events; discarding such mostly irrelevant events helps manage measurement error in the antitrust indicator. Finally, we normalize the estimate by the total market capitalization of listed firms and aggregate at a quarterly or annual frequency.

Civil non-merger and criminal cases | Merger cases

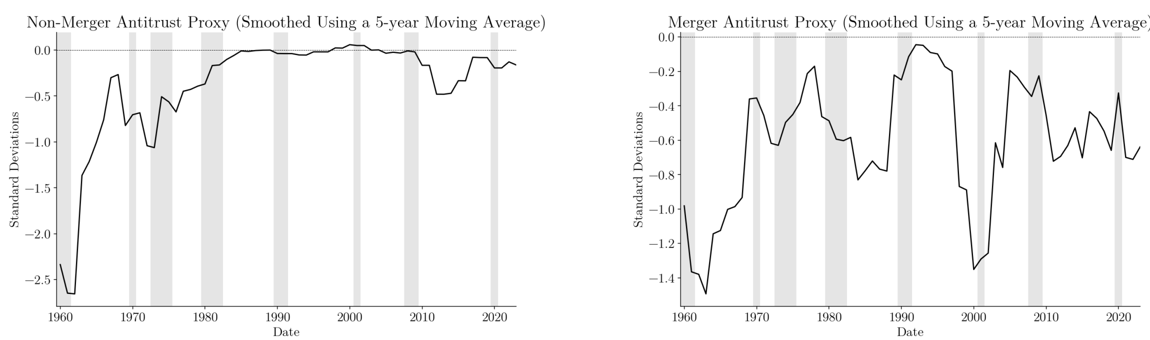


Figure 8: Estimated Indicator for Antitrust Activity

In the analysis below, we will use this indicator of antitrust activity evaluated at the two-digit industry level. However, it is still of interest to inspect its path over time at the aggregate level. Figure 8 illustrates a five-year moving average of the indicator of antitrust activity for civil non-merger and criminal cases and for merger cases. We indicate NBER recessions by the grey-shaded areas. Notably, despite the fact that equity returns are cyclical, we do not find any strong evidence of cyclicality in the antitrust activity measures.

For civil non-merger and criminal cases, we find three waves of high antitrust activity. The first wave runs from the late 1950s to the early 1960s, during the Dwight D. Eisenhower and John F. Kennedy administrations. Measured in terms of standard deviations, this is the period of most intense antitrust activity against publicly listed firms in the sample.

For instance, in October 1958, the USDOJ launched a case against nine oil companies and independent marketers of gasoline for fixing the retail price of gasoline. In December 1958, the USDOJ launched a case against several of the successors to the Standard Oil Company for engaging in territorial allocation and tying arrangements regarding the sale of refined oil products. In 1960, the USDOJ initiated multiple cases against dozens of manufacturers of electrical equipment for price fixing and bid rigging (notably, General Electric was listed as a defendant in 31 different criminal and civil non-merger cases during that year). The second wave runs from the late 1960s to the mid-1970s, during the Lyndon B. Johnson and Richard Nixon administrations. Relative to later periods, this period also stands out as having intense antitrust activity. Notably, in 1972, the USDOJ launched a case involving intellectual-property abuses that included 17 publicly listed companies from the defense industry as defendants.

A 15-year period of antitrust tranquility began in the early 1980s and lasted until the mid-1990s, when the USDOJ launched well-known investigations such as the case against Nasdaq market makers for having colluded on quoting conventions (launched in July 1996), the case against Akzo Nobel Chemicals and Glucona for price fixing and market allocation for sodium gluconate (launched in September 1997), and cases against VISA International Corp. and Mastercard International Inc. (launched in October 1998) for restraints on competition in the market for general-purpose credit-card products and services. We do not, however, find a large impact of these cases in terms of abnormal returns. Instead, we find some impact of cases raised for such violations in the early 2010s under the Barack Obama administration, when the USDOJ started antitrust investigations against Apple Inc. and a large group of commercial publishers for conspiring to limit competition in e-book markets (in April 2012), and against Verizon and a group of other cable-TV suppliers for bundling offerings in violation of Section 1 of the Sherman Act.

The behavior of the civil-merger-case antitrust indicator is very different. First, while this indicator displays considerable variation over time, it shows no signs of a structural decline or increase in merger-related antitrust activity. As with the indicator for civil non-merger and criminal cases, we find high antitrust activity from the late 1950s to the mid-1960s. A second wave of high intensity occurs in the mid-1980s, during the Reagan administration, and a third in the late 1990s, during the Clinton administration. Overall, this antitrust activity indicator for merger-related cases brought against public firms changes much less over the sample period than the indicator for civil non-merger and criminal cases against public firms.

4 Estimating Markups

Below we estimate the impact of antitrust activity on sector-level and aggregate outcomes. One outcome that we examine is the gross markup, defined as price divided by marginal cost. We follow much recent work and estimate markups for listed firms using a production-function approach applied to Compustat accounting data. Our focus on Compustat is natural because the estimates of antitrust activity discussed above are based on equity returns.

Following Hall (1988), De Loecker and Warzynski (2012), De Loecker et al. (2020), and many other recent papers, we exploit cost minimization to express the gross markup as follows:

$$\mu_{j,t} = \frac{\theta^v_{j,t}}{\alpha^v_{j,t}} \tag{7}$$

$$\theta^v_{j,t} = \frac{\partial Q_{j,t}}{\partial X^v_{j,t}} \frac{X^v_{j,t}}{Q_{j,t}} \tag{8}$$

$$\alpha^v_{j,t} = \frac{P^v_{j,t} X^v_{j,t}}{P_{j,t} Q_{j,t}} \tag{9}$$

where $\mu_{j,t} = P_{j,t}/MC_{j,t}$ is firm j's gross markup at date t, $\theta^v_{j,t}$ is the firm's output elasticity with respect to the variable input $X^v_{j,t}$, and $\alpha^v_{j,t}$ is the variable input's share of revenue. The associated Lerner index is $1 - 1/\mu_{j,t}$. Finally, $Q_{j,t}$ denotes the quantity of firm j's output, $P_{j,t}$ is its product price, and $P^v_{j,t}$ is the variable input price. When reporting markups in percentage terms below, we report $100(\mu_{j,t} - 1)$.

Subject to assumptions about which factors are variable, the factor spending share, $\alpha^v_{j,t}$, is observable in accounting data such as Compustat. The output elasticity with respect to the variable input instead needs to be estimated. For this purpose, we use the Ackerberg et al. (2015) two-step version of the Olley and Pakes (1996) estimator. We assume that production technologies are identical within two-digit NAICS industries and specify them using either translog or Cobb-Douglas technologies:

$$q_{j,t} = \theta_{0,l} + \theta_{1,l}x^d_{j,t} + \theta_{2,l}x^v_{j,t} + \theta_{3,l}(x^d_{j,t})^2 + \theta_{4,l}(x^v_{j,t})^2 + \theta_{5,l}x^d_{j,t}x^v_{j,t} + z^p_{j,t} + z^{tr}_{j,t}, \quad j \in N_l \tag{10}$$

$$q_{j,t} = \theta_{0,l} + \theta_{1,l}x^d_{j,t} + \theta_{2,l}x^v_{j,t} + z^p_{j,t} + z^{tr}_{j,t}, \quad j \in N_l \tag{11}$$

where $q_{j,t}$ denotes the logarithm of firm j's output at date t, $x^d_{j,t}$ denotes the logarithm of the dynamic input, $x^v_{j,t}$ denotes the logarithm of the variable (static) input, $z^p_{j,t}$ is a persistent productivity factor, and $z^{tr}_{j,t}$ is a non-persistent productivity factor. N_l is a set of firm indices that determines the industry membership, l, of firm j.

The persistent productivity factor is assumed to be first-order Markovian with a distribution that is known to firms and stochastically increasing in $z^p_{j,t}$, while $\mathbb{E}_{j,t}z^{tr}_{j,t} = 0$. It is assumed that there are no news shocks to the persistent productivity component. Along with the literature, we will assume that the dynamic input is "capital" which is accumulated over time by combining existing capital, $x^d_{j,t-1}$, with "investment" at date $t-1$, $i_{j,t-1}$. Both inputs are assumed to be chosen prior to the realization of $z^{tr}_{j,t}$. The key distinction between the capital and variable (or static) factors of production is that the former are chosen at date $t-1$ prior to the firm having any information about $z^p_{j,t}$, while the latter may be chosen anytime from period $t-1$ up until just prior to the realization of $z^{tr}_{j,t}$.

With these assumptions, following Ackerberg et al. (2015), the input elasticities can be estimated in two stages using the control-function approach of Olley and Pakes (1996). We apply the control function to the variable input, which is assumed to be increasing in $z^p_{j,t}$. Following De Loecker et al. (2020), we allow the control function to depend on market shares to address the "omitted price problem" that arises because product prices are not available in Compustat data. De Ridder et al. (2026) study the seriousness of the omitted price problem and argue that appropriate choices of the econometric framework, including ours, still allow for meaningful estimates of trends in markups and markup dispersion.

The estimation allows factor prices to vary over time at the sector level but excludes persistent firm-specific factor-price components. It allows for capital adjustment costs only insofar as they are not firm-specific. We measure variable costs by cost of goods sold (COGS), which includes spending on labor and raw materials, as well as items such as manufacturing overheads. We estimate the production function using annual data for the sample period 1954 to 2023 and use standard selection criteria; see, e.g., Hasenzagl and Pérez (2026). Our baseline estimates use the translog specification because of its flexibility in approximating production functions; see also De Ridder et al. (2026). As we discuss later, none of our results are very sensitive to instead assuming a Cobb-Douglas technology.

Our analysis below will study markup estimates at the sector level. In order to aggregate markups across firms, we compute sales-weighted harmonic means:

$$\mu_{l,t} = \left(\sum_{j \in N_l} \frac{P_{j,t} Q_{j,t}}{P_{l,t} Q_{l,t}} \frac{1}{\mu_{j,t}} \right)^{-1} \tag{12}$$

where $P_{j,t}Q_{j,t}$ is firm j's sales and $P_{l,t}Q_{l,t}$ is total sales of firms in the sector. Likewise, we aggregate to the total Compustat level using sales-weighted sector-level harmonic means. Arguments for using the sales-weighted harmonic mean as a sectoral or aggregate markup measure can be found in Grassi (2018), Baqaee and Farhi (2020), Edmond et al. (2023) and Hasenzagl and Pérez (2026); this measure follows from aggregating gross markups across firms.[14]

Figure 9 illustrates our estimates of the aggregate markup assuming either translog production functions (Panel A) or Cobb-Douglas production functions (Panel B). The levels of the estimated aggregate markups and their time paths are very similar for the two alternative specifications. For the translog specification, we find a 15-percentage-point increase in the aggregate markup over the sample, from around 10 percent at the beginning to around 25 percent in the 2020s, while for the Cobb-Douglas specification, it rises by 18 percentage points, from 15 percent at the beginning to around 33 percent by the 2020s. The two estimates also agree on the variation over time, both displaying a rise in markups from the early 1960s to the early 1970s, followed by a marked decline from the 1970s to the early 1980s. From the early 1980s, as has been much discussed, aggregate markups display strong and sustained growth for two decades, rising from levels of 5 percent (12 percent) in the early 1980s to 24 percent (33 percent) by the mid-2000s for the translog (Cobb-Douglas) specification.

[14] Equivalently, sector-level markups can be computed as employment-weighted arithmetic means; see Edmond et al. (2023).

Translog Production Function | Cobb-Douglas Production Function

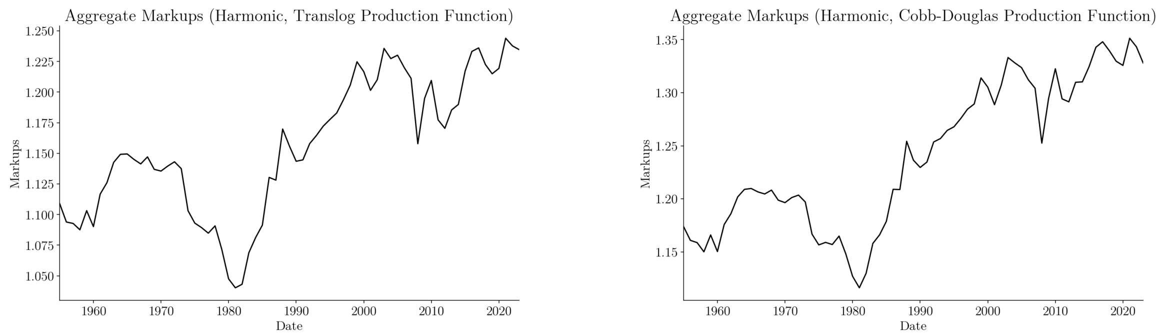


Figure 9: Aggregate Markup Estimates

Translog Production Function | Cobb-Douglas Production Function

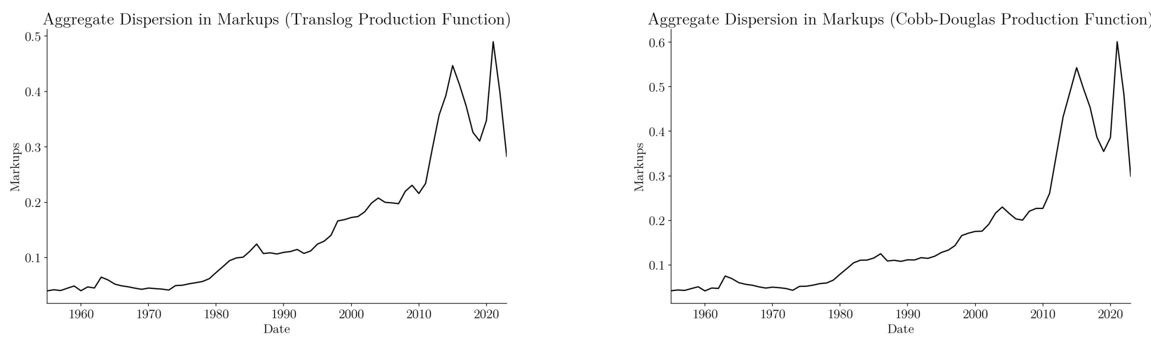


Figure 10: Markup Dispersion

Our aggregate markup estimates agree with those of Hasenzagl and Pérez (2026), who impose Cobb–Douglas technologies but use investment when constructing the control function. However, the rise in the aggregate markup that we estimate is much smaller than that in De Loecker et al. (2020), who find a rise of more than 40 percentage points over the same period. As discussed in detail by Hasenzagl and Pérez (2026), the main difference stems from the use of harmonic means when aggregating. When we instead use arithmetic averages, our estimates are similar to those of De Loecker et al. (2020).

In Figure 10, we illustrate the dispersion of markups across firms. Regardless of the production function specification, we find a strong rise in the dispersion of markups across Compustat firms that starts in the late 1970s/early 1980s, continues until around 2010, accelerates significantly until 2015, and becomes very volatile thereafter. In models with linear pricing, this rise in dispersion can be interpreted as increasing misallocation. In such environments, markup dispersion is a sufficient statistic for factor misallocation because firms with high markups sell "too little" relative to the case in which factors are efficiently allocated across firms; see, e.g., Baqaee and Farhi (2020), Edmond et al. (2023).[15]

5 The Dynamic Impact of Antitrust Activity: Sector-Level Evidence

5.1 Approach

We now exploit the indicators of antitrust activity constructed in Section 3 to estimate the dynamic impact of antitrust. We initially focus on sector-level evidence and aggregate the firm-level estimates to the two-digit sector level. With these indicators in hand, the dynamic effects of antitrust are estimated using a panel local-projection (LP) approach; see, e.g., Dube et al. (2025).

The estimating relationships are specified as:

$$y_{h,t+s} - y_{h,t-1} = \delta_{t,s} + \psi_{h,s} + \beta_s \triangle \nu_{h,t} + \mathbf{\Gamma}_s \triangle X_{h,t-1} + e_{h,t+s} \tag{13}$$

where $y_{h,t} \in X_{h,t}$ denotes the outcome in question for sector $h \in H$, and $\nu_{h,t}$ is the estimated impact of antitrust cases against firms in sector h at date t. In order to control for other shocks, we allow for an aggregate time-fixed effect, $\delta_{t,s}$, for sector-specific effects through $\psi_{h,s}$, and for a rich set of controls. We specify the latter by controlling for lags of the entire set of outcomes that we consider. Because of differencing, sector-fixed effects do not appear in the panel LP regressions but are present in the implicit levels specifications. We compute standard errors with the Driscoll and Kraay (1998) method using 8 lags. Moreover, due to high persistence of the data, we correct for the Nickell bias using a split-panel jackknife estimator; see Dhaene and Jochmans (2015).

We focus on two-digit BEA/NAICS sector-level aggregation. The data are annual and cover the sample period 1957–2023. We examine the dynamic impact on the following vector of outcomes: real GDP and price levels obtained from the BEA; operating profits measured from Compustat; the sales share of the treated sector, measured as total sales of Compustat

[15] Bornstein and Peter (2025) show that this result may be sensitive to non-linear pricing which induces misallocation across consumers.

firms in the respective sector relative to total sales of Compustat firms; two-digit concentration, measured by the Herfindahl index; markups and markup dispersion, as estimated in Section 4; and TFPR. All variables are measured in natural logarithms. We measure TFPR from the production-function estimates in Section 4 and aggregate within sectors. In the baseline regressions, we use the translog production-function estimates for markups and productivity, but we also show results for the Cobb-Douglas specification. We include various indicators of competition in the analysis because focusing on a single measure is potentially misleading. As discussed by, e.g., Syverson (2019), market concentration is an outcome determined by the conditions of competition, and a high level of concentration may not necessarily indicate a less competitive environment. Markups are, in principle, a better indicator but are harder to measure. By examining a set of indicators, we hope to draw firmer conclusions about how antitrust affects market power and competition.

5.2 Results

Figure 11 illustrates the dynamic responses of indicators of competition and market power to the antitrust indicator related to civil non-merger and criminal cases. These variables are operating profits, markups, and the Herfindahl index. We show the point estimates for a forecast horizon of up to five years, along with 68 percent and 90 percent confidence bands indicated by the dark- and light-shaded areas, respectively.

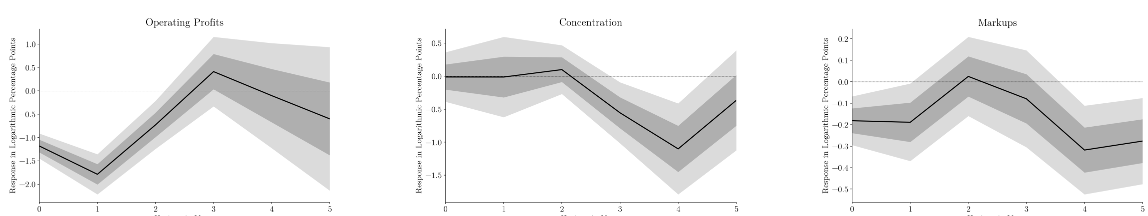


Figure 11: The Dynamic Impact of Civil Non-Merger and Criminal Cases on Competition

We find that when a sector is subject to more intense antitrust activity, the operating profits of the Compustat firms in the sector fall abruptly upon impact and remain depressed for an additional two years, after which they recover. Accompanying the decline in profits, we also find a persistent decline in markups which is significant at the 90 percent level upon impact and in the first, fourth and fifth years thereafter. The signs of increased competition are also reflected in the Herfindahl index but occur with some delay. Thus, on the basis of these indicators, we find that more intense antitrust activity against civil non-merger and criminal offenses induces a decline in market power of firms in the treated sector relative to other sectors.

Figure 12 illustrates the impact of the antitrust measure on other outcomes. Most significantly, we find an abrupt decline in the Compustat firms' capital expenditures, which takes place upon impact and lasts for the first two years thereafter. Moreover, although there is some recovery of capital expenditures in the third and fourth years, the results indicate a significant decline in investment. Alongside the decline in investment, we also estimate an increase in markup dispersion in the treated sector, which persists for four years and is significant at the 68 percent level upon impact and in the first and fourth years. The decline in capital expenditures and the increase in markup dispersion are accompanied by a persistent decline in the sector's BEA real value added, which is significant at the 68 percent level for the first four years. The decline in real value added is not reflected in the sales share of the Compustat firms in the sector except in the first year. This may indicate either an impact on costs or that part of the transmission occurs through non-listed firms. In favor of the cost channel, our results indicate a decline in TFPR until four years after the treatment. Finally, we find a temporary reduction in prices upon impact, which is reversed at longer forecast horizons.

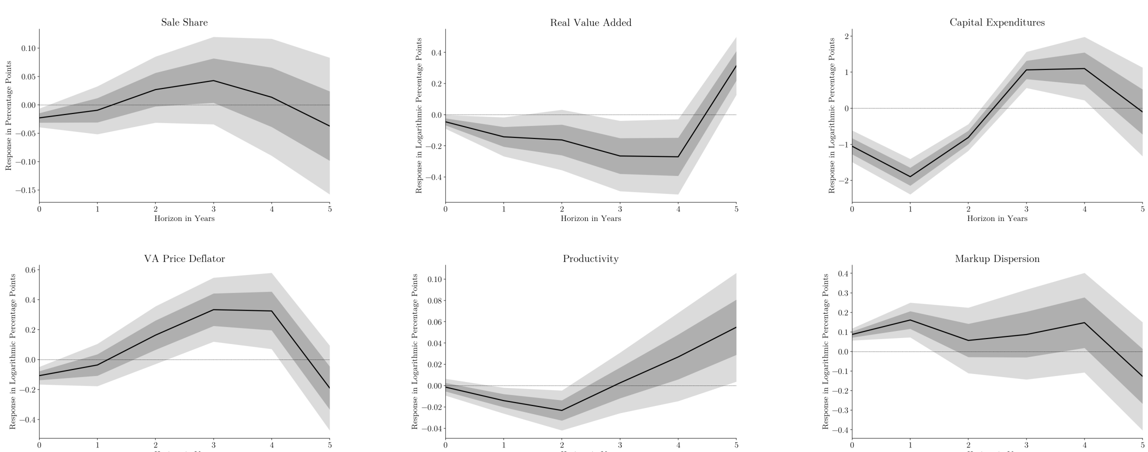


Figure 12: The Dynamic Impact of Civil Non-Merger and Criminal Cases on Sector-Level Outcomes

These results indicate that although increased antitrust activity related to civil non-merger and criminal violations of antitrust laws is associated with declining market power, there is little evidence of beneficial effects on the economy in the short to medium run. Importantly, our results indicate detrimental effects in terms of capital investment, misallocation, and productivity. Interestingly, Werden (2014) argues that the rule of reason focuses solely on competition rather than on efficiency and welfare. Nonetheless, given that our estimates are ultimately based on cross-industry comparisons, it is unclear whether the results mask the impact of spillovers on other industries (i.e., whether there is a "missing intercept" issue). Below, we examine aggregate effects to study these issues in more detail.

Figure 13: The Dynamic Impact of Civil Merger Cases on Competition Measures

Figure 13 illustrates the dynamic impact of the merger-case-based antitrust measure on the market-power indicators. Consistent with the results for civil non-merger and criminal cases, we find that firm concentration and markups decline persistently in the treated sector relative to other sectors. The impact on markups occurs more gradually over time than for civil non-merger and criminal cases but also appears more persistent. The decline in the Herfindahl index is less precisely estimated than for civil non-merger and criminal cases but is significant at the 68 percent level at the three-year horizon. We find no significant impact of merger-related antitrust on operating profits, which, if anything, rise. Nonetheless, the fact that firm concentration and markups both decline makes the results broadly consistent with merger cases stimulating competition. In this respect, one must also account for the fact that many M&A cases may affect competition measures over the longer run. These effects are harder to estimate precisely.

The wider sector-level impacts of the civil-merger antitrust indicator are shown in Figure 14. In contrast to the results for civil non-merger and criminal cases, we find that the prevention of mergers stimulates capital expenditures by firms in the treated sector relative to other sectors. The rise in capital expenditures is estimated to be both large and persistent. Moreover, more intense merger-related antitrust activity is associated with a decline in markup dispersion, which is statistically significant at the 90 percent level in the third, fourth, and fifth years. These results are in stark contrast to those reported earlier for civil non-merger and criminal cases. More in line with the results in Figure 12, we find a negative impact on real value added, but it is quantitatively small and significant only at the 68 percent level at the three-year forecast horizon. In the case of mergers, a key consideration is always the trade-off between the productivity and market-power effects of M&A. Thus, it is perhaps not surprising that we find that relative sector-level TFPR declines in the short to medium run following an increase in the civil-merger-related antitrust measure.

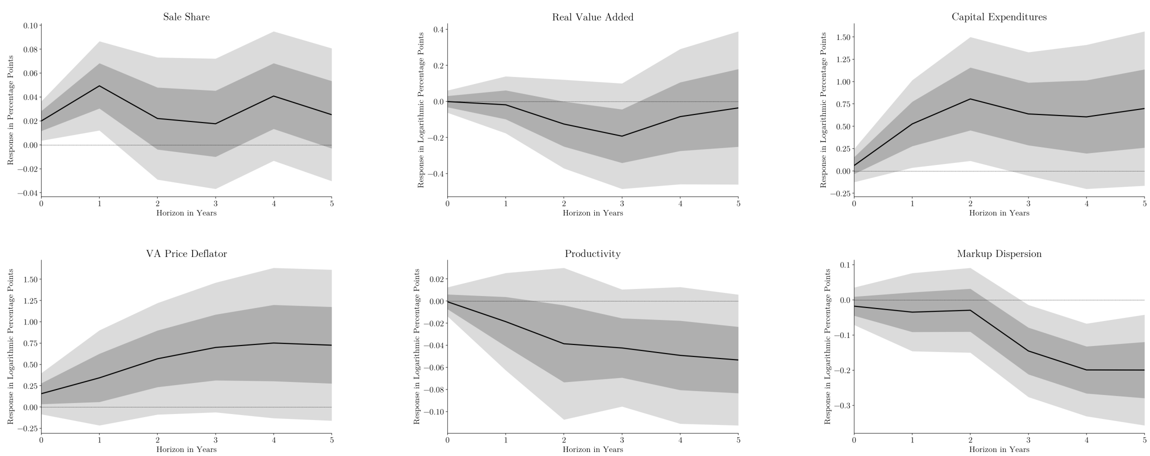


Figure 14: The Dynamic Impact of Civil Merger Cases on Sector-Level Outcomes

5.3 Robustness

It is important to establish the robustness of our results to various measurement-related issues.

5.3.1 The Technology Specification

A first issue concerns the measurement of productivity and markups. The results reported above build on translog production-function estimates as in equation (10). The translog specification has the advantage of allowing differences in output elasticities across firms, sectors, and time. On the other hand, because of this flexibility, the translog specification may also induce instability in estimates of markups and productivity at the firm level. For that reason, we now examine robustness to estimating markups and TFPR from a Cobb–Douglas production-function specification as in equation (11), imposing parameter homogeneity at the two-digit level.

We estimate the elasticity of output to variable inputs (θ_v), the elasticity to dynamic inputs (θ_d), and the implied returns to scale (ξ) at the two-digit NAICS level for both the translog and Cobb–Douglas production-function specifications. For the translog case, the coefficients are evaluated at the mean factor inputs. The estimates are very similar across the two technologies, apart from construction (sector 23), where the Cobb-Douglas specification implies a smaller variable-input share than the translog case, and FIRE (sector 52) and education services (61), where the variable-input elasticities are estimated to be higher for the Cobb-Douglas specification. However, the parameter estimates are very similar in the vast majority of cases.

Figure 15 illustrates the estimates of the dynamic impact of antitrust on markups, markup dispersion, and TFPR when assuming a Cobb–Douglas technology. Given the evidence discussed above, it is not surprising that the results are very similar to those discussed earlier for the translog case: higher antitrust activity leads to a decline in markups, while the impact on markup dispersion depends on the antitrust violations, with dispersion declining in the case of M&A-type violations and rising for civil non-merger and criminal cases. The impact on TFPR for M&A violations is also very similar to the estimates from the translog case. The only instability in the results concerns the impact on TFPR for civil non-merger and criminal cases. For this variable, the translog case indicates a decline over the first three years, while the Cobb-Douglas case implies a rise in TFPR. TFPR is inherently difficult to estimate and interpret, and this result suggests that these responses must be interpreted with due care. For all other outcomes discussed under the baseline specification, the dynamic responses are very similar for the Cobb-Douglas technology.

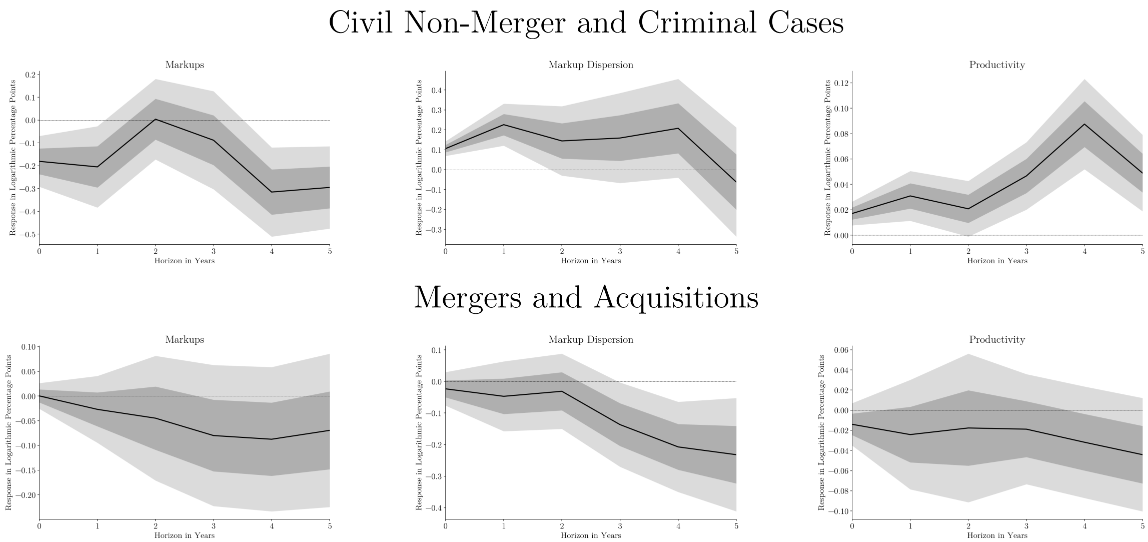


Figure 15: Cobb-Douglas Production Function

5.3.2 Excluding FIRE

A second concern with our results is that we have included FIRE in the baseline estimates. Many analyses involving estimates of markups or productivity based on the production-function approach exclude FIRE because of problems estimating production functions, and therefore markups and TFPR, for this sector.

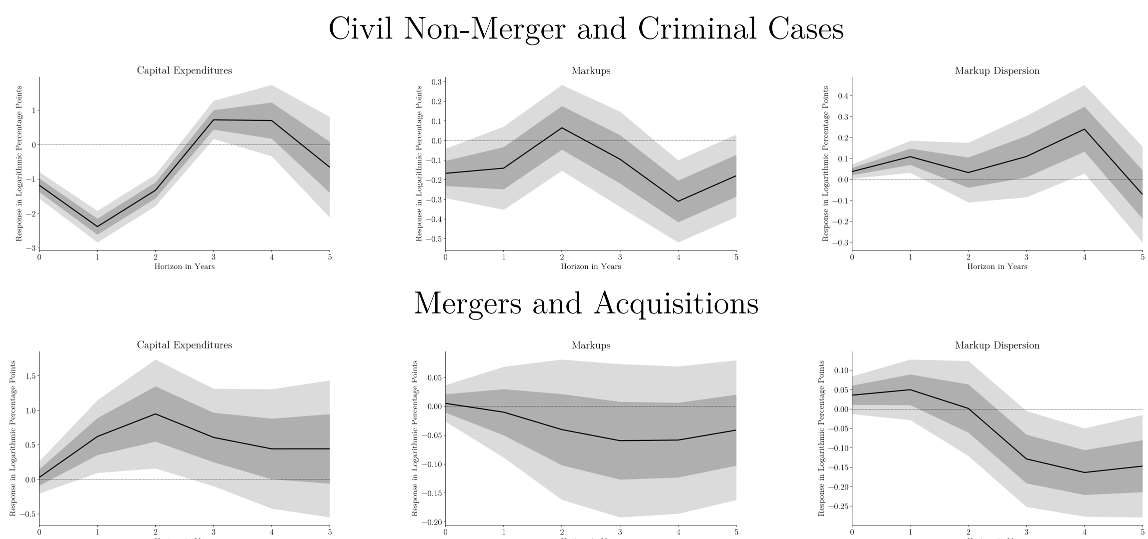


Figure 16: Excluding FIRE

In Figure 16, we report the impact of antitrust activity on three key outcomes—capital expenditures, markups, and markup dispersion—when we exclude FIRE from the analysis. The responses of these variables are very similar to the baseline estimates, except that the decline in markup dispersion appears with a delay of two years for civil mergers and acquisitions. The results are also robust for all other outcomes.

The results are also almost identical when utilities are excluded from the analysis.

5.3.3 Excluding the Early Sample

A third consideration relates to the sample period. As discussed in Section 2.4, the period from the late 1950s to the early 1960s witnessed unusually high levels of antitrust activity, especially on the part of the FTC. One might be concerned that our results are therefore driven by this part of the sample period and are not representative of the whole sample.

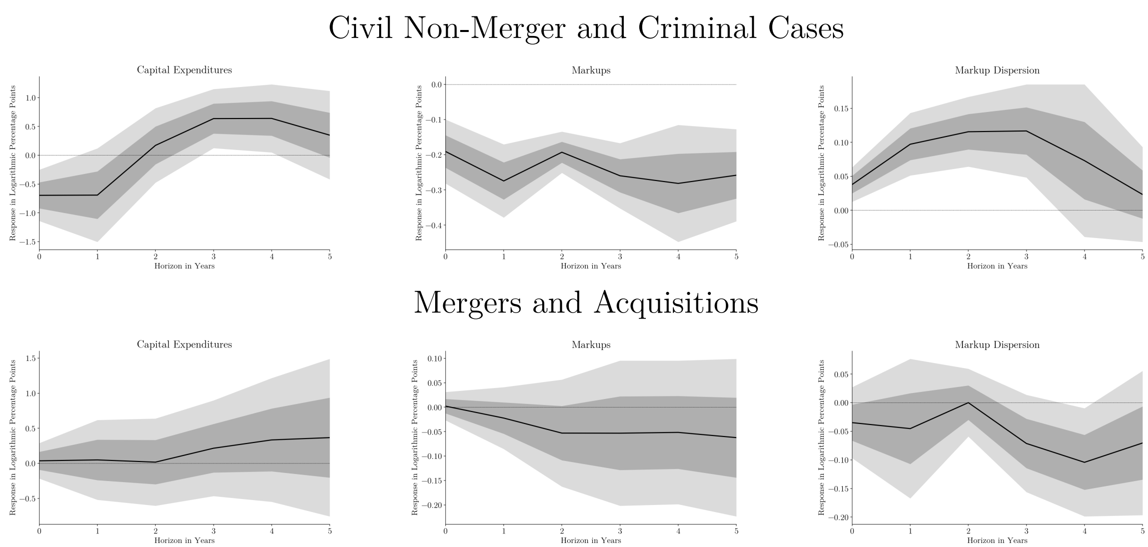


Figure 17: Estimates for the 1962–2023 Sample Period

We illustrate the results for the key variables when starting the sample in 1962 in Figure 17. Again, we find that the key results are robust, although there is some change in the magnitude of the response of capital expenditures and a more pronounced decline in markups in response to the antitrust indicator for civil non-merger and criminal cases. For mergers and acquisitions, we similarly find a more muted rise in capital expenditures. For all other outcomes, the results are virtually unchanged.

We have also examined robustness along a number of other dimensions, such as eliminating the rich set of controls, not allowing for sector-specific trends, evaluating the cumulated excess returns at shorter or longer horizons when building the antitrust activity indicators, and changing the cutoff level for the cumulated excess returns when constructing the antitrust indicators. All the results are also robust across these dimensions.

6 Aggregate Results

The results presented in the previous section suggest that the two types of antitrust cases both stimulate competition but have contrasting impacts on sector-level outcomes. This leaves open the question of their aggregate effects because of the "missing intercept" problem arising from general equilibrium effects and spillovers. Thus, we now turn to estimates of the dynamic macroeconomic effects by aggregating the antitrust indicators across sectors and exploring how they affect aggregate U.S. economic outcomes.

For this purpose, we estimate local projections of the form:

$$y_{t+h} = \alpha_h + \delta_h t + \beta_h \nu_t + \Gamma_h X_{t-1} + u_{t+h} \tag{14}$$

where ν_t is the antitrust indicator aggregated across firms and sectors.

We estimate the local projections for a rich set of outcomes. As above, we examine the impact on markups, markup dispersion, and TFPR estimated from the translog specification of the technology. To this, we add BEA data on real profits, real GDP, real consumption, real investment, real R&D spending, the unemployment rate, real hourly earnings, and labor productivity. We also examine the impact on the federal funds rate, CPI inflation, and real stock prices. All variables apart from the unemployment rate and the federal funds rate are measured in natural logarithms. When estimating an outcome for a variable y, we include two lags of this variable among the controls in X_{t-1}. Furthermore, in the vector of controls, we include two lags of markups, real GDP, inflation, the federal funds rate, real stock prices, R&D, real earnings, the unemployment rate, and real corporate profits. We apply the bias correction for short time series by Herbst and Johannsen (2024), and we compute Newey-West standard errors using eight lags.

6.1 Civil Non-Merger and Criminal Cases

Figure 18 reports the responses of markups, real profits, and markup dispersion to the civil non-merger and criminal-case antitrust indicator. We show the point estimates along with 68 and 90 percent confidence bands. The results echo those that we found in the sector-level analysis: an increase in antitrust activity depresses real profits and gives rise to a persistent decline in markups, while markup dispersion rises. Thus, in line with the results in Section 5, we find signs of increased competition but also more misallocation, indicating an economy-wide impact beyond the sector-level results discussed earlier.

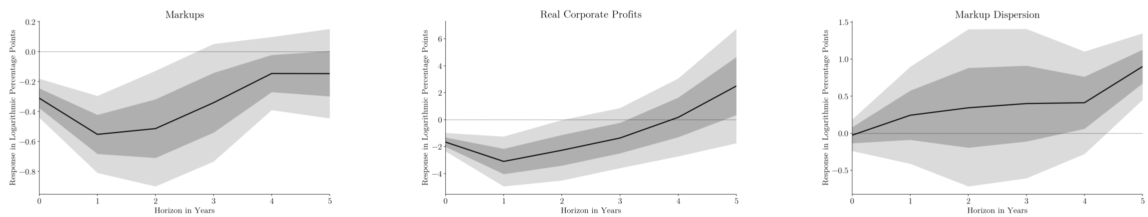


Figure 18: The Dynamic Impact of Civil Non-Merger and Criminal Cases on Competition Measures

In Figure 19, we report the results for real outcomes. The results indicate persistent negative effects of civil non-merger and criminal-case antitrust activity on the economy. In particular, we find persistent and significant declines in real GDP, consumption, investment, and R&D, accompanied by declining labor earnings and a rise in unemployment. The negative effects are particularly pronounced for real investment and R&D spending and show no signs of recovery within the forecast horizon that we examine. Unemployment is the exception, showing signs of recovery after four years. Furthermore, we find a persistent decline in TFPR, while labor productivity hardly moves.

Finally, Figure 20 shows the impact of the antitrust indicator on the federal funds rate, CPI inflation, and real stock prices. The results indicate that the Federal Reserve leans against the wind and reduces the short-term nominal interest rate in response to the impact of the antitrust indicator on the economy. Hence, there are conflicting pressures on inflation, given the decline in markups and the real economy combined with a monetary stimulus. We find the net effect to be a temporary drop in inflation. We also find that real stock prices decline, indicating that the decline in real interest rates does not offset the impact on

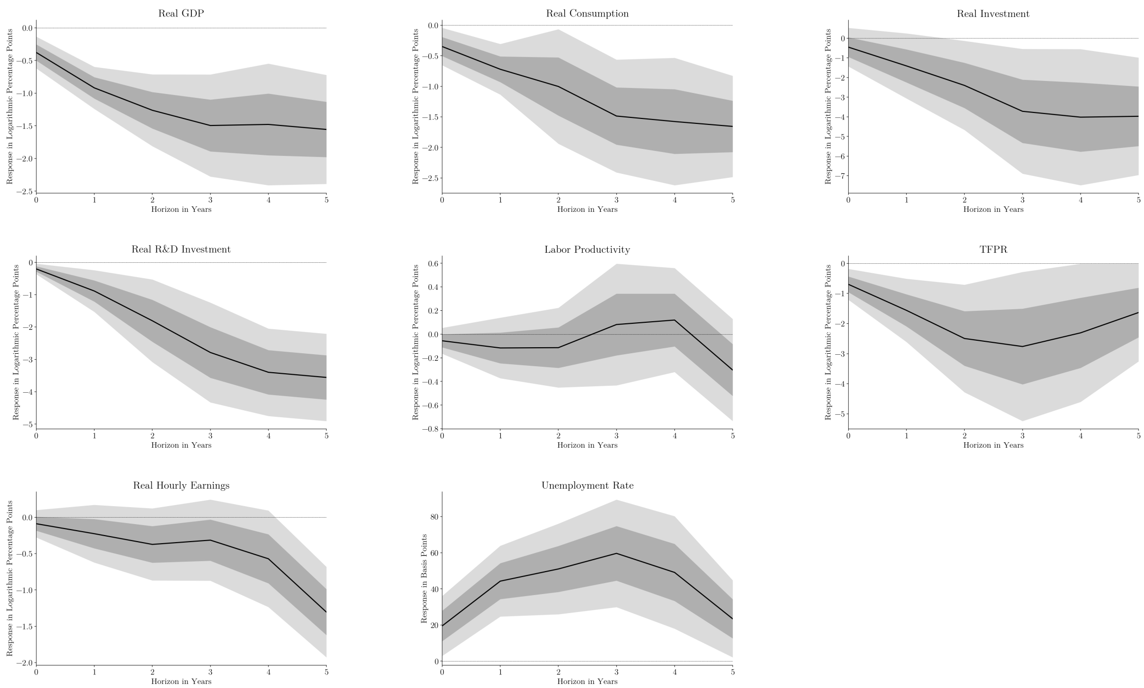


Figure 19: The Dynamic Impact of Civil Non-Merger and Criminal Cases on the Real Economy

expected stock returns.

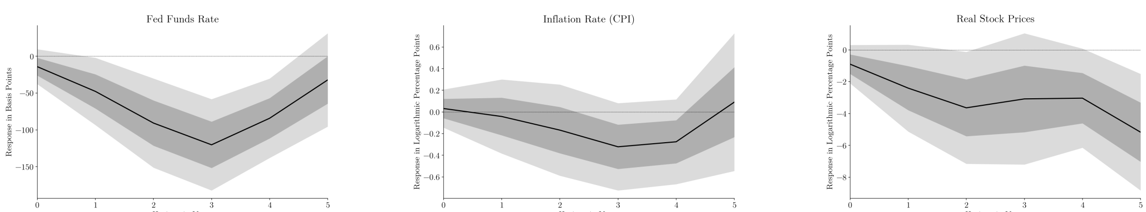


Figure 20: The Dynamic Impact of Civil Non-Merger and Criminal Cases on Asset Markets and Inflation

To sum up, our results indicate that more intense antitrust activity in terms of civil non-merger and criminal cases raised against publicly traded firms depresses market power, but at the cost of a negative impact on the economy. We have investigated the robustness of these results to the measurement of markups and TFPR using the results from the Cobb-Douglas specification of the production function, to the elimination of the time trend in the LP regressions, and to the measurement of the antitrust indicator using either different forecast horizons for the cumulated excess returns or different filters for the minimum impact on abnormal returns. We find that the results are robust, with the sole exception, as discussed earlier for the sector-level results, that the TFPR response is sensitive to the specification of the production function: when assuming a Cobb-Douglas technology, we find only a mild decline in TFPR.

6.2 Civil Mergers and Acquisitions

Next, we turn to the aggregate impact of civil-merger antitrust investigations. Figure 21 shows the impulse responses of markups, corporate-sector profits, and markup dispersion to the merger-based antitrust indicator. We find that more intense antitrust activity induces a decline in markups and corporate-sector profits. Both variables respond with some delay and with a lower elasticity than under the civil non-merger and criminal-case indicator but are more indicative of pro-competitive effects than the corresponding sector-level estimates. As far as markup dispersion is concerned, we find some decline in the short run, which reverses at longer forecast horizons, but standard errors are large and sampling uncertainty does not permit us to reach a clear conclusion on misallocation.

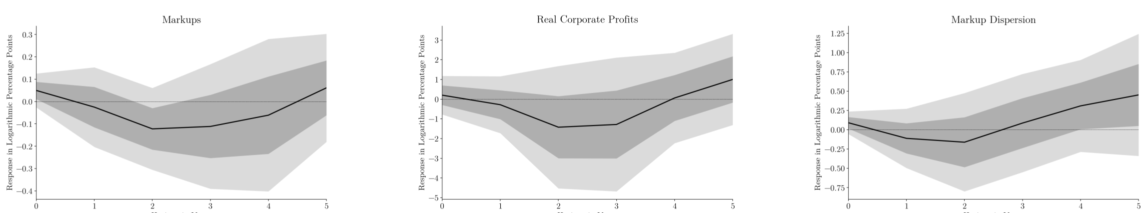


Figure 21: The Dynamic Impact of Civil Merger Cases on Competition Measures

The impact on the real economy is illustrated in Figure 22. We find that more intense antitrust merger enforcement has a stimulatory impact on the economy. In particular, we find a rise in real GDP, which is significant on impact and two (three) years thereafter at the 90 percent (68 percent) level. Accompanying the rise in real GDP, we find that private-sector consumption, real investment, and real R&D spending all rise. The short-run impact on investment is large but imprecisely estimated, while the rise in consumption is more moderate but more precisely estimated, with significance at the 90 percent level for three years.

We also find that real hourly earnings and labor productivity respond positively to an increase in the merger-case-related antitrust indicator, while unemployment declines, albeit

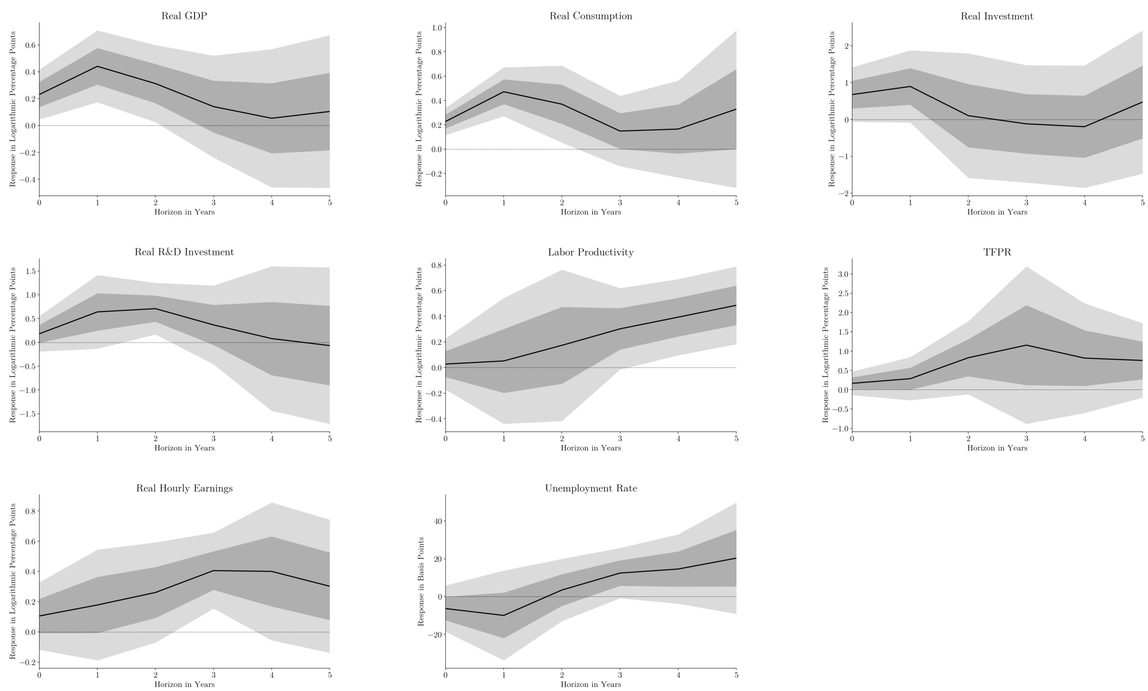


Figure 22: The Dynamic Impact of Civil Merger Cases on the Real Economy

only temporarily. Finally, we find an increase in TFPR, but sampling uncertainty is high for this variable.

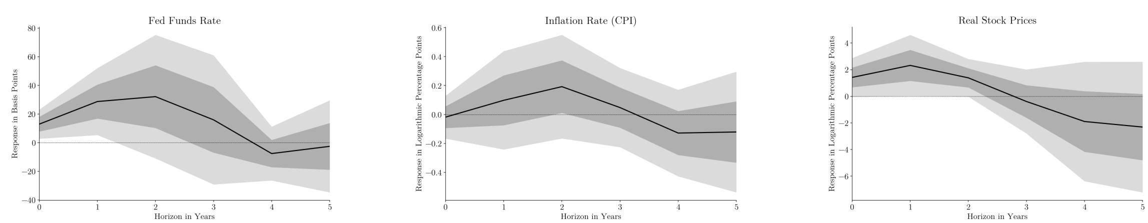


Figure 23: The Dynamic Impact of Civil Merger Cases on Asset Markets and Inflation

Finally, in Figure 23, we show the impact on the federal funds rate, CPI inflation, and real stock prices. Most likely due to the impact of antitrust activity on aggregate activity, we find that the federal funds rate and CPI inflation rise marginally in response to more intense civil-merger-related antitrust activity for the first 3–4 years, after which the effect dies out. In terms of the stock market, we estimate a positive effect on firm valuations on impact and in the first two years thereafter, suggesting that preventing mergers has beneficial effects on the valuations of other firms in the economy.

The contrast between the results estimated for civil non-mergers and criminal cases and those for civil merger cases is striking. Our results suggest that prevention of anti-competitive mergers has benefited the economy, while we fail to uncover such evidence for civil non-merger and criminal cases. These results also indicate that the change in antitrust activity around 1980—which involved a stark decline in the intensity of indictments for civil non-merger and criminal cases and an increased focus on mergers—may have been beneficial in protecting the economy against the harmful aspects of firm concentration.

6.3 Quarterly Results

The results discussed so far are all based on annual data. All other things equal, it would be preferable to study quarterly data since this gives us many more observations and can address issues related to time-aggregation. The main issue here is that the markup, markup dispersion, and TFPR estimates are derived from annual Compustat data. Quarterly Compustat data are only available for a much shorter sample and for a subset of the variables for a subset of the firms. As an alternative, we now consider proxying markups by the inverse of the labor share, which permits one to generate quarterly estimates of the impact of antitrust. This approach has costs because it does not allow us to control for markup dispersion or to study market concentration. Furthermore, the (inverse) labor share only proxies for the Lerner index under special circumstances, so our results should be interpreted with caution.

We show the results in Figure 24 for a forecast horizon going up to 20 quarters. The qualitative results for the civil non-merger and criminal cases are very similar to the annual results discussed above. We find that the labor share increases in response to more intense antitrust activity, which is consistent with the drop in markups discussed above. In the annual data, we found that the drop in markups persists for four years, and in the quarterly data, we find that the rise in the labor share persists for 15 quarters. As in the annual data, we also find a negative and very persistent impact on real GDP, consumption, investment, and R&D spending. The main difference relative to the annual results is that we fail to find a significant impact on real hourly earnings. Nonetheless, the quarterly results are remarkably similar to those we found in the annual data. For civil merger cases, we instead fail to find an impact on the labor share. This may indicate that the labor share is a poor proxy for the markup in this case and/or that the lack of controls for markup dispersion and market concentration impairs the estimation. For this reason, the quarterly results are hard to interpret in this case. Interestingly, though, there is still a stark contrast to the results for the civil non-merger and criminal cases because we find a positive impact of the civil merger antitrust indicator on labor productivity and real hourly earnings, while the estimated responses of real output, investment, and R&D spending are not significantly different from zero.

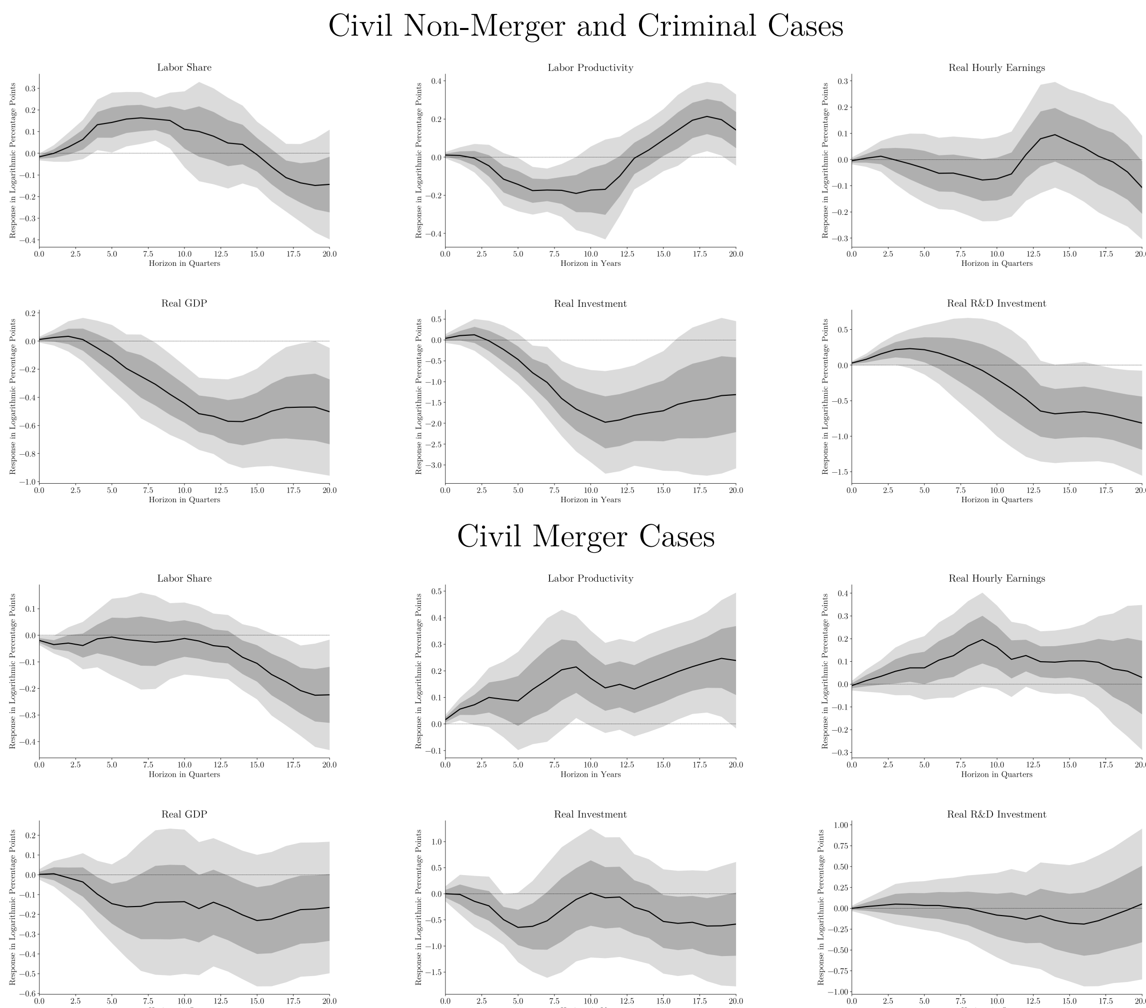


Figure 24: Quarterly Results

7 Conclusion

In this paper, we have examined the link between antitrust activity, competition, and aggregate outcomes. We collected data on U.S. federal antitrust investigations since the mid-1950s. We documented a fundamental shift in the focus of antitrust away from civil non-merger and criminal cases towards civil-merger-related violations of antitrust laws, a shift that is particularly evident for publicly listed firms. We also documented that publicly listed firms indicted for antitrust violations are heavily selected from the right tail of the firm-size distribution even relative to other publicly listed firms.

We map corporate firms involved in these cases to Compustat data and estimate how firm valuations are affected by antitrust activity. We demonstrate that, on average, firms that are indicted for antitrust violations suffer significant negative abnormal stock returns after the antitrust cases are opened. The abnormal returns are larger for civil-merger-related violations than for civil non-merger and criminal cases. We exploit the firm-level estimates to build indicators of antitrust activity measured as the impact on firm valuations within a certain sector or across the economy. These indicators show meaningful variation over time and across sectors, which allows us to estimate their impact on the economy.

We find that when a sector is more intensively subjected to antitrust activity, indicators of market power decline. However, the impact on other outcomes differs markedly across types of antitrust violations. In particular, we find that increased antitrust activity related to civil-merger violations stimulates economic activity, including capital investment and R&D spending, and raises labor productivity and real wages, while increased antitrust activity related to civil non-merger and criminal cases has the opposite effects. Consistent with this, we also find that markup dispersion declines in the former case but increases for civil non-mergers and criminal cases.

One interpretation of our results is that antitrust activity related to civil non-merger and criminal cases has potentially been counterproductive because, while stimulating competition, it has had negative consequences for the economy. To the extent that one can consider the antitrust activity indicator a proxy for changes in competition, the results are also consistent with the fact that certain types of changes in market power may be good for the economy.

Given that this type of antitrust activity relates mainly to the pre-1980 sample, these results are consistent with that period being subject to forces that simultaneously stimulated the economy and increased market power. In contrast, preventing anti-competitive mergers appears to stimulate both competition and the overall economy. Thus, the rise in merger activity that occurred around 1980 may have been associated with "bad concentration" which the antitrust authorities attempted to lean against. Nonetheless, given the nature of our analysis, we leave a full analysis of this to further research. These interpretations would be consistent with the "good" vs. "bad" concentration hypothesis of Covarrubias et al. (2020), perhaps combined with issues related to the implementation of antitrust laws in terms of the relative importance assigned to competition and efficiency considerations.

Investigating the underlying structural interpretation of the results in more detail is clearly important and is the focus of our ongoing research. In this respect, Amorim Cabaço (2025) examines the impact of mergers on the economy using a structural firm-dynamics model and argues that the anti-competitive effects of mergers dominate the pro-productivity effects. It would follow from this that the change in antitrust focus that we have documented in the data has been warranted, but perhaps also that enforcement has been too lax to prevent rising market power.

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